Chinese Journal of Mechanical Engineering 39 (2026) 100032

Contents lists available at ScienceDirect

Chinese Journal of Mechanical Engineering

journal homepage: www.keaipublishing.com/en/journals/cjme/



Recent progress and challenges of key technologies in robotic assembly

Longhui Qin[a,b,*]

[a] School of Mechanical Engineering, Southeast University, Nanjing 211189, China
[b] State Key Laboratory of Mechanical Transmission for Advanced Equipment, Chongqing University, Chongqing 400044, China

ARTICLE INFO

Keywords:
Robotic assembly
Perception
End-effector
Skills learning
Learning from demonstration
Learning from interaction

ABSTRACT

How to substitute the human operator with a robot in various assembly tasks has to be taken into full consideration in intelligent manufacturing. Autonomous robotic assembly not only brings with high working efficiency, better product quality and low labor cost, but also helps relieve the increasingly severe problem of population aging. However, numerous existing challenges still prevent its wide applications when a robot is assigned to finish general tasks in unstructured environment. In order to provide a fundamental understanding of the various problems involved in robotic assembly, this paper carries out a review on its recent progress and challenges with 5 key technologies focused on: perception, end-effectors, control methods, learning methods and performance evaluation. Main works in these fields are reviewed and their characteristics are analyzed while typical assembly scenarios are covered. The challenges and future directions in robotic assembly are also discussed on precise perception, robotic hand, error recovery and collaborative robot. In addition to providing a systematic summarization of the required key technologies, this work is aimed at motivating more potential researches in the community of robotics, artificial intelligence, and automation engineering.

1. Introduction

As a common procedure in everyday industrial production, assembly of mechanical parts affects the efficiency, precision and quality of final products. However, the conventional assembly manner that is implemented by human operators is suffering from inevitable problems and challenges resulted from the enormous increase of population aging and labor cost. In recent years, more and more researchers turn their attentions to robotic assembly, i.e., substituting the conventional human operators with intelligent robots in assembly tasks. A variety of investigations on robotic assembly can be found in plenty of applications, such as 3 C production, architecture construction, medical surgery, aerospace manufacturing and so on [1–4].

Different from most existing applications in the community of robotics, robotic assembly in unstructured environments is still challenging and not widely applied nowadays since lots of difficulties remain unsolved and hamper the improvement of uncertainty, precision and generalization in practical applications. As a result, current robotic assembly is usually realized either in structured environment, i.e., working conditions are fixed and the motions and actions are predefined, or assisted by human operators via frequent human-robot interaction (HRI) or remote manipulation. Numerous endeavors have been devoted to the autonomous implementation of typical assembly tasks, e.g., peg-in-hole (PiH), cable routing, bolt tightening, connector mating and screw driving, which are facilitated by recent advances in the community of artificial intelligence and robotics. The purpose of this review is to introduce the main works and advancements in autonomous robotic assembly, summarize the critical techniques, and discuss potential research directions.

A robot that is able to implement autonomous assembly is expected to equip with the ability of interacting with surrounding environment and tackling unexpected variant situations. The involved key techniques are described in Fig. 1. Properties and statuses of the objects to assemble as well as the proprioceptive information of the robot is indispensable in contact-rich scenarios. According to the employed sensors, current perception manners mainly include visual, force/torque (F/T), and tactile perception. Meanwhile, multiple types can be fused to pursuit higher precision. End-effector (EE) refers to the mechanical structure installed at the end tip of a robotic arm, with which various manipulation actions, e.g., grasping, placement, insertion and release, can be implemented. In addition to the most widely used EE, grippers, robotic hands that provide more degree-of-freedom (DoF) and dexterity show great potential in flexible assembly while task-specified EEs have been validated in many applications. Control models play a critical role

* Corresponding author at: School of Mechanical Engineering, Southeast University, Nanjing 211189, China.
E-mail address: lhqin@seu.edu.cn.

https://doi.org/10.1016/j.cjme.2025.100032
Received 25 November 2024; Received in revised form 2 June 2025; Accepted 18 August 2025
Available online 1 September 2025
1000-9345/© 2025 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

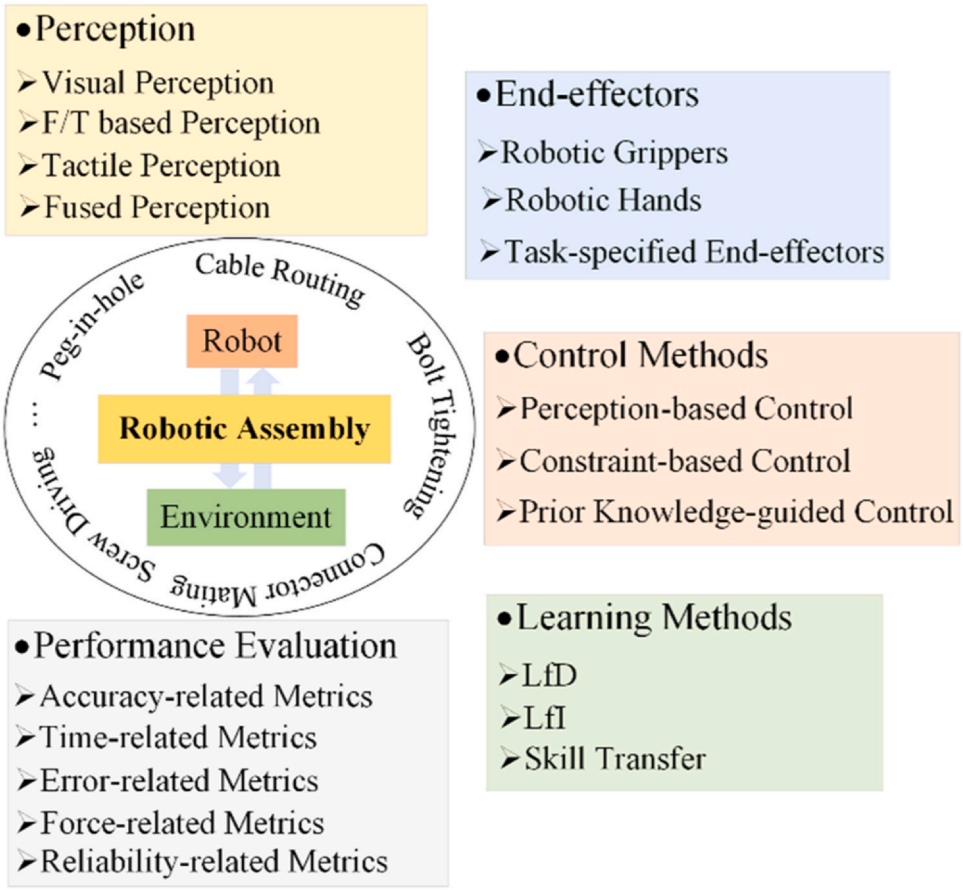


Fig. 1. Involved key components in robotic assembly.

in robotic assembly and it determines what actions should be taken next according to the current states and feedback. As per their dominating basis, existing control methods can be categorized into perception-based, constraint-based and prior knowledge-guided control. How to learn the assembly skills from human or interactions and transfer them to a robot has received enormous attentions from various research fields these years. Learning from demonstration (LfD) and learning from interaction (LfI) are two prevalent subfields, and the skill transfer strategy aims to fill the gap between simulation and reality. To assess the performance of an assembly strategy, a series of experiments will be conducted, and abundant evaluation metrics from different perspectives have been proposed. In addition, it should be noted that we limit ourselves to the automatic assembly with serial-chain manipulators among various robotic fields in this paper.

As multifarious issues are involved in robotic assembly, existing reviews have tried to address a part of critical problems. The methods and strategies for robotic grasping and assembly tasks are reviewed in Ref. [5], which focused on the utilization of sensing and constraints in robotic manipulation. In Ref. [6], state-of-art search strategies were described with PiH task focused on, where the role of different sensors and data fusion methods were discussed to realize target search precisely. A more comprehensive review, covering the issues of target recognition and searching, assembly strategies, fault detection and performance evaluation, was provided in Ref. [7], which mainly focused on the PiH task. In addition, the bolt assembly and wire harness assembly scenarios were discussed on their present status and future trends while the difficulty and complexity definitions for task allocation and assignment were explored to facilitate human-robot collaborations [8–10]. Different from existing reviews, this paper tries to extract and analyze the essential hardware and software components in multiple mainstream robotic assembly applications. While not limiting ourselves within a specific assembly task, this review elucidates a complete pipeline of robotic assembly with the state-of-art vital techniques introduced. Furthermore, the challenging problems and future directions are discussed on precise perception, dexterous hand, error recovery and collaborative robot.

The rest of this review is organized as follows: Section 2 presents the prevalent perception manners for assembly parts and contact status, followed by the introduction of widely employed EEs in Section 4. Control and learning methods in robotic assembly are respectively depicted in Sections 4 and 5. After the summarization of four types of evaluation metrics Section 7, this review is concluded with further discussions on potential directions in Section 7.

2. Perception in robotic assembly

In unstructured environment, a mass of information is required in advance of decision making and next action implementation, which usually serves as an essential component in control and learning strategies. The quantities to perceive can be classified into two types: (1) Properties and states of the manipulated objects, such as the shape and size, the pose (position and orientation), and the motion trajectory and velocity. (2) Assembly status. It includes not only the contact forces and torques, but also the proprioceptive configuration of the robot itself. The prevalent perception manners include visual, F/T based, tactile and fused perception as per the employed sensors.

2.1. Visual perception

According to the installation location of visual sensors and whether they are fixed or moving, there are three widely adopted ways in robotic assembly as shown in Fig. 2: eye-in-hand, eye-to-hand, and their

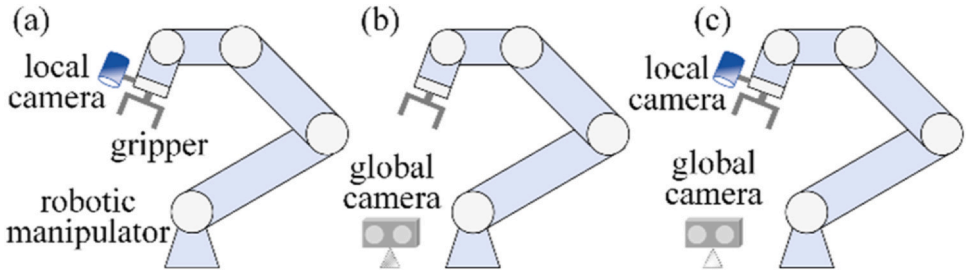


Fig. 2. Three manners of visual perception: (a) Eye-in-hand manner, (b) Eye-to-hand manner, (c) Hybrid manner.

hybrid manner. In eye-in-hand manner, the camera is attached near the EE, and it moves and captures multiple local images of the region of interest (ROI) from different views and distances. As for eye-to-hand, camera is fixed at a predefined location, maybe at the robot base, upside of the worktable, or other locations that enable global captures of the whole manipulation space. A hybrid manner of them is able to combine their advantages by obtaining both global and local ROI images. In visual perception, there exist many types of sensors and various algorithms have been proposed.

2.1.1. 2D visual sensors

Five typical sensors in robotic assembly are illustrated in Fig. 3 [11,12]. Monocular camera acts as one of the widely used visual sensors (Fig. 3(a)), which captures the two-dimensional (2D) images and advantageous in its low-cost and easy utilization. Usually, an eye-in-hand manner is adopted and the robotic arm cooperates with the camera by move-and-capture actions at different time instants. An estimation of the 3D trajectory motion of assembly point on a moving workpiece was realized by a developed robot-guided monocular vision system [13]. In the assembly of a cylinder block in motion, it was found the trajectory and 6D pose could be accurately reconstructed via the triangulation with at least two views. A step-by-step method was proposed in Ref. [14] with a dual-arm robot Baxter for object positioning and grasping in PiH task. The ROI images from different views and distances help providing more details. In the application of transformer insertion, the features of transformer pin images (vision layer) were utilized to estimate the insertion motions (motion layer), which was inputted into an intelligent model to generate ideal insertion motions (decision layer) [15]. In order to mitigate the accumulation errors caused by the lengthy coordinate chain propagation, a monocular visual measurement method was developed for hole-to-hole and PiH tasks based on local pose invariance and 3D-2D matching [16]. A camera adjustment method was proposed based on a rough location of the target screw, which improved the measurement accuracy in the disassembly of hexagonal screws [17].

2.1.2. 3D visual sensors

When a pair of monocular cameras is combined, their captured parallaxed images constitute the binocular vision, which enables stereopsis and additionally obtains the depth information. An automatic system was developed for the assembly of smartphone back shells using the binocular camera, consisting of look-then-move control for distal assembly and look-and-move control for proximal assembly [18]. By exploiting the reprojection of 3D edges onto binocular RGB images, the 6D pose can be accurately estimated with initial pose estimation and refinement via edge-map-based nonlinear optimization [19]. Two monocular cameras were installed at different locations near the EE to capture images from different views, which were inputted into separate CNNs for feature extraction and then concatenated to feed into a reinforcement learning (RL) model [20]. It validated the feasibility of robust robotic insertion of electronic component.

A more popular sensor to acquire 3D visions is depth camera as shown in Fig. 3(b), and according to the detection principles there are stereo, time-of-flight, light detection ranging, and structured light. A RGB-D sensor was applied to estimate the success probability of assembly state with the aid of Bayesian classification [21]. 3D CAD (Computer Aided Design) models were used to train a deep convolutional neural network (CNN), aimed at realizing detection, pose

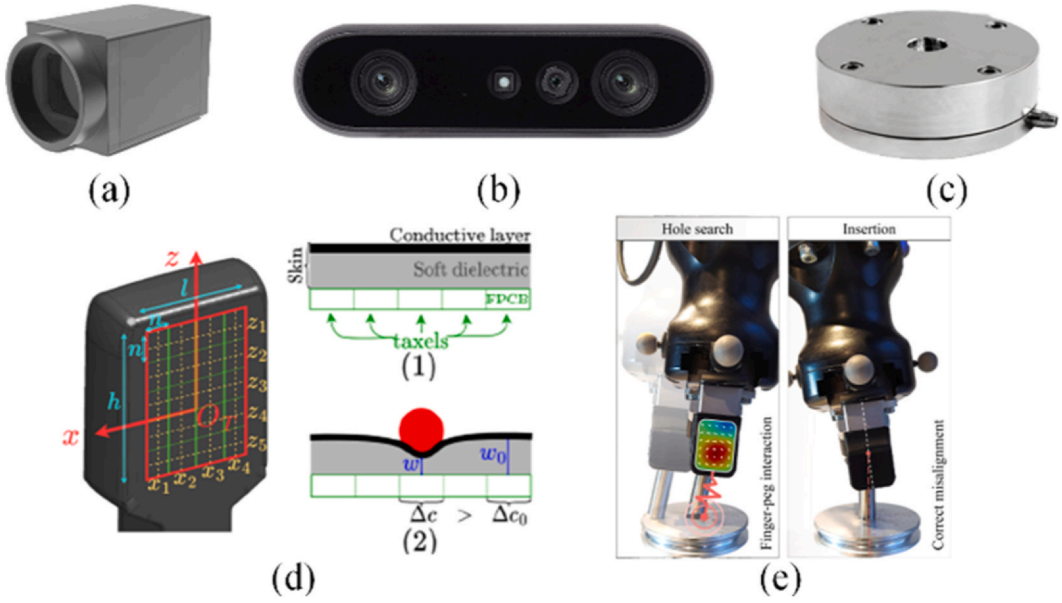


Fig. 3. Five typical sensors in robotic assembly: (a) 2D camera, (b) Depth camera, (c) F/T sensor, (d) Capacitive tactile array in Ref. [11], (e) Vision-based tactile sensor for PiH task in Ref. [12].

estimation, refinement and handling of complex parts, which was found effective in simulating the depth images and facilitating millimeter-accuracy [22]. To solve the cable routing problem with a dual-arm robot, a depth camera was fixed at the front of the robot, i.e., eye-to-hand manner, responsible for poses estimation of the fixtures and tracking of the deformable linear object [23].

2.1.3. Hybrid visual sensors

To obtain higher perception accuracy, a combination of eye-in-hand and eye-to hand manners is preferred in practical applications while both monocular camera and depth sensor are deployed, i.e., hybrid visual manner. It not only increases the operation flexibility of the EE, but also improves the perception effectiveness. In the assembly of mobile phone which included the main board, the screen and the back shell, the eye-to-hand industrial camera was employed to detect the assembly target, positioning the workpiece and guiding the robot to target position. Meanwhile, the eye-in-hand camera captured images of ROI and then recognized the assembly area via template matching [24]. A combined two-level vision system was developed in Ref. [25] and Ref. [26] for PiH tasks. In the former investigation, high precision was obtained in the measurement of local hole pose and global robot pose. The pose of eye-in-hand camera was adaptively adjusted based on rough pose measurement by the eye-to-hand binocular camera in the latter research.

2.1.4. Visual perception algorithms

Among existing visual perceptions in robotic assembly, three perception objectives are intensively investigated: *object recognition*, *pose estimation*, and *positioning*.

Object recognition is the most fundamental step in robotic assembly and its result directly affects the assembly performance. In Ref. [27], the CAD model facilitated to detect the threaded hole via edge detection applying the difference in gradient method, and the bolt was detected using the reference surface model and surface matching technique. An uncertainty-driven spiral trajectory guided by exploration distribution was proposed to search the hole in Ref. [28]. Dual monocular cameras were integrated to find the flat key and keyway for robotic assembly of keyed circular peg via deep learning (DL) based image semantic segmentation [29]. YOLOv4 was adopted to detect screw targets and EfficientNetv2 was applied for screw shape classification in robotic disassembly [30].

Pose estimation is required before the assembly actions in most scenarios and enormous efforts have been devoted to how to raise its accuracy. As for RGB-only objects, physically-based rending helped in the 6D pose estimation [31]. While prior knowledge, e.g., the layout feature of connector, was helpful [32], plenty of studies took full advantages of the interaction process, such as moving the camera to search for the best viewpoints [33], physical interacting with the environment via 'look' and 'touch' actions in combination with F/T sensor [34], and implementing interactive perception motions mimicking human behavior [35].

Positioning aims to determine the relative position between different parts to assemble, which provides key information for alignment and visual servoing control. A measurement method based on high-precision fiducial markers was proposed to realize positioning based visual servoing in microrobotic assembly, in which a positioning accuracy below 50 nm was achieved [36]. The positioning error was affected by the hand-eye configuration, which was discussed in Ref. [37] from a perspective of tolerance management. Two cameras were arranged orthogonally during the grasping and installation stages while a point feature of the object to assemble was extracted to construct image-based visual servoing for position alignment [38].

Visual perception in robotic assembly differs from that of other systems in that the local camera can move flexibly at very low time and hardware cost. Therefore, inspired by the exploration way of human operators, the perception algorithm can be coordinated with the robot motion, realizing a combination of passive and active perception.

2.2. F/T based perception

In the contact-rich environment, robotic assembly requires accurate estimation of the involved forces and torques in order to assess the current contact status and determine next actions.

2.2.1. F/T sensors

Force/Torque (F/T) sensors are utilized to perceive the forces and torques applied from external objects or environment. One of its most prevalent type is the 6-axis sensor (Fig. 3(c)), or called multi-axis F/T transducer, capable of measuring the three force components, F_x, F_y, F_z, and the three torques, T_x, T_y, T_z. Among various sensing technologies, resistive strain gauges are widely employed to develop F/T sensors due to its cost-effective and reliable performances. In most researches on robotic assembly, commercial F/T sensors are adopted directly [39–42], and there also exist a few investigations on the design and improvement of novel F/T sensors [43]. These sensors are usually installed between the EE and end tip of the robotic arm. In such a way, the force and torque applied to the EE can be measured more precisely in contact-rich tasks. Unusual installation way could also be found in a few investigations, e.g., the F/T sensor was fixed below a hole and parallel to the robot base, based on which the pose precision between the peg and hole was improved, and a piecewise strategy and decoupling control method was proposed [42].

2.2.2. F/T perception

With the F/T sensor, different assembly states can be recognized, such as contact status (CS), anomaly, and motion transients. In the assembly of low-voltage circuit breaker, five topical contact states were recognized based on extreme learning machine (ELM) as the 6-axis F/T components served as input layer [44]. The CS was investigated by modeling the wrench signal (Cartesian force and torque) in PiH assembly for flexible rubber objects using Gaussian mixture model (GMM) [45,46]. Utilizing the F/T sensor behind the connect mount on the robot, a Gaussian process regression (GPR)-based model was established to learn the force profile during successful insertions and detect anomalies, achieving an accuracy of 100% in differentiating between normal and faulty insertions [47]. Transients in measured F/T data were detected to realize force-controlled assembly of the internals of an emergency stop button [39]. By recording the F/T data applied by the operator and feeding them to a CNN, automatic insertion was implemented with the impedance control-based trajectory teaching [40]. In addition, the F/T sensor plays a significant role in numerous control models since the forces and torques it provides are the basis to construct F/T-based controller in all assembly scenarios [48–50], which will be introduced detailedly in Section 5.1.

2.2.3. Proprioceptive perception

While F/T sensors are employed as external devices in most works, there emerge a subfield emphasizing inexpensive sensorless control without external sensors [51–54]. Instead, the internal torque sensors at each joint of a robot are taken advantage of, also called *proprioceptive perception*. To implement PiH assembly, a model-based sensorless observer of interaction was introduced via the joint torque sensors equipped by the robot, which could estimate and reconstruct external torques [51]. Considering the contact force affected each joint and generated external torques, a PiH strategy consisting of four unit motions 'pushing', 'rubbing', 'wiggling' and 'screwing' was proposed without force feedback [52]. A disturbance observer based on generalized momentum and joint motor torque was proposed to estimate the force information on the end tip of a manipulator with the algorithm calibrated via a CNN supervised learning method [53]. Only the robot's proprioceptive sensory information was required in force-based robotic insertion, where a novel active search strategy was formulated exploiting partially observable Markov decision process (MDP) [54].

Although no external sensors are employed in proprioceptive perception leading to a cost reduction, the chain of information transmission becomes longer and thus, the reconstruction and estimation of assembly states is more difficult from the EE end in task space to the joint space. Therefore, F/T sensor stays predominant in existing assembly tasks.

2.3. Tactile perception

Compared with the above two perception manners, the application of tactile sensors to robotic assembly is still in the budding stage but has shown significant potential considering it enables the in-hand manipulation and more resembles the way a human operator works.

2.3.1. Traditional tactile sensors

Artificial skin belongs to a typical tactile sensor and has received a great deal of investigations these years. In Ref. [55], artificial skin covering the manipulation handle acted as an user-interface to recognize human voluntary interactions in collaborative assembly tasks. Tactile array, if fabricated as a thin layer, can be attached to robotic EEs for assembly applications. In order to blindly pick small screws from bins through in-finger manipulation, a sensor array comprising 6×4 sensing elements (SEs) was pasted on a fingertip with thin sponge layer attached to it, which obtained an average success rate of 80% using RL [56]. When a tactile sensor composed of 25 photo-reflectors (5×5 matrix) was attached to a gripper to manipulate boltlike fasteners, it was successfully validated that the tactile perception facilitated the robot to adapt different objects and shapes under uncertain/perturbed conditions [57]. In cable routing scenario, tactile data from a capacitive sensor (Fig. 3(d)) could be exploited to reconstruct the shape of the grasped portion of deformable cables [11].

2.3.2. Vision-based tactile sensors

Vision-based tactile sensors emerge as a novel type in recent years. It mainly consists of micro camera and lighting devices underneath transparent soft layers, obtaining the tactile information via captured images of deformed markers. A visual tactile sensor was developed in Ref. [58] comprising two layers of transparent elastomer, markers, camera, and light source, with which a bolt was inserted into the target hole and then tightened. As shown in Fig. 3(e), A vision-based tactile sensor, Digit, was leveraged for anomaly detection and hole search using a variational autoencoder model and model-based learning control [12]. With a fingertip sensor GelSight, tactile-guided low-level motion control and vison-based high-level task parsing method were proposed in cable routing and the assembly of a reconfigurable task board [59].

With tactile sensors, complexity of the control model can also be reduced as tactile sensors directly contact the objects to assemble, providing direct measurement of the object properties and CS. As a result, estimation of the influence of external environments and contact on the EE and the robot joints is no longer required. What's more, the perception becomes more flexible since the tactile sensors are usually arranged at the fingertip of robotic gripper and hand, which allows active multi-point exploration.

2.3.3. Future directions of tactile perception in robotic assembly

In recent years, remarkable progress has been achieved in the development of tactile sensors and corresponding signal processing algorithms. Although a vast number of investigations focus the application of tactile sensors to wearable electronics, human motion monitoring, and gait analysis etc., there are also lots of works related to robotic assembly, such as tactile-based grasping, where target recognition, pose estimation, and force measurement are involved in Refs. [60,61]. Among various working principles, four primary types of tactile sensors are being widely used in robotic grasping, i.e., capacitive, piezoelectric, piezoresistive and visual-tactile sensors. Recent advancements are gained in both the design of tactile sensors and the development of sensory data processing methods. While the former includes material innovation, integration of multiple functionalities, design method improvement and manufacture process optimization, the latter advances mainly in how to apply ML and DL techniques so as to enhance the perception performance [60,62,63]. Tactile perception has been proven effective in target recognition, pose estimation, force measurement, state discrimination and destabilization adjustment, which are closely related to the contact-rich assembly scenes [61]. Meanwhile, tactile data generation is vital in the performance of visuo-tactile sensors, where tactile images are generated and DL algorithms that require large size of training data are usually leveraged. The physical simulation techniques, e.g., elastomer simulation and optical simulation, had progressed these years and learning-based methods were thought more suitable for collecting robot manipulation data [64]. Robotic manipulation and assembly with visuo-tactile sensors could deeply explored in future.

2.4. Fused perception

The performance and capability of single perception manner appears limited, resulting in increased complexity of the perception algorithms and control models. Fused perception, a combination of several different sensing technologies, can alleviate such burdens to a great extent. How to fuse different sensors in complex robotic scenarios to enhance the assembly efficiency and accuracy deserves in-depth investigations. There are mainly two fusion mechanisms underlying the fused perception. For the first type, each sensor is used to perceive distinct objectives or sense the same objective at different stages or under different situations, and the sensory data is proposed by respective methods or models. In such a manner, the combination of multiple sensors helps improve the adaptability and flexibility of assembly strategies, and reduce human interventions. In another type, the data collected from all sensors are proposed by the same perception model at a time. It belongs to multimodal perception when different kinds of sensors are employed. Throughout existing investigations, it can be found that most works of fused perception fall into the first type.

Visual-F/T-fused perception is the most prevalent fusion manner, as vision provides recognition function in non-contact situations and F/T dominates in contact scenes, which belongs to the first fusion type. Two depth cameras, one fixed to the environment and the other one attached to the robot wrist, were used to detect graspable points on the parts and check successful grasp, while a F/T sensor was applied to advance the robot until a physical contact occurred and move the bearing shaft screw forward during insertion process [65]. With this fused perception, each sensor worked in its preponderant situation and thus, the precision of every procedure could be ensured. Based on three microscopic cameras and a micro F/T sensor, a three-stage `aligning-approaching-grasping' control strategy was proposed [66]. While the images captured by two cameras were processed to extract the component edges via Random Sample Consensus (RANSAC) method, the measurement data from force sensor was used to adjust the assembly component via threshold comparison. In PiH assembly, a depth camera captured the relative position between the peg and hole, and a six-axis F/T sensor was employed to estimate the external forces and moments [67]. In reaching phase, the vision information facilitated the realization of visual-based policy by applying a convolutional neural network (CNN) model to extract image features and the force measurement was leveraged in searching phase to construct constant-force control.

Meanwhile, robotic vision and tactile sensing can also be integrated. In the application of wire insertion on electromechanical components, collision was detected by exploiting tactile feedback in case of wrong alignment and wire terminal pose was corrected using vision feedback [68]. Therein, the magnitude of collision force was predicted via a MLP (multilayer perception) regressor with tactile sensor, and the wire terminal was detected with a self-training CNN. As other types of

sensors were employed in robotic assembly, e.g., acoustic transducer, optical scanner, and laser tracker, they can also be fused to enhance the performance [69–71]. Two cameras and three laser distance sensors (LDS) were employed for automatic guidance and alignment in the assembly of large-scale objects [72]. Although camera-2 and 3 LDS were simultaneously utilized in the stage of close-range alignment, adjustment based on vision sensor was implemented first and then it was followed by the adjustment based on LDS. Along with the development of various sensing technologies, fused perception from more sensors and modalities may appear for the purpose of higher precision and stronger robustness.

The typical perceptions in robotic assembly are listed in Table 1.

3. End-effectors in robotic assembly

EEs, as the executive element, are responsible for action implementation in robotic assembly. In addition to the positioning accuracy, there are other factors affecting the execution effect in practical assembly, such as the number of fingers, degree of freedom (DoF), actuation mechanism, and adaptability etc. In this section, the EEs in robotic assembly will be introduced in three categories: robotic grippers, robotic hands and task-specified EEs. Here, the robotic hand refers to the EE that more resembles human hand in the appearance and execution manner and it usually comprises more multi-joint fingers. In Fig. 4, five typical EEs in robotic assembly are exemplified [73–76].

3.1. Robotic grippers

Robotic gripper is one of the most widely used EEs and plenty of commercial grippers are available in the market. Generally speaking, robotic gripper is equipped with very limited DoF (1 or 2 DoF in most cases) and often two or three fingers. As shown in Fig. 4(a), both of them possess only one DoF and a close or open action can be implemented by the two parallel fingers.

To empower the gripper with more functionalities in assembly tasks, various designs of novel grippers have been proposed. By combining two shape adaptive mechanisms, i.e., granular jamming and single wire-driven multi-finger, a novel gripper was designed to adaptively grasp objects of different shapes, which comprised two active fingers and two passive ones (Fig. 4(b)) [73]. A soft adaptive assembly module was proposed in Ref. [77] to reduce the cost in assembly processes, allowing for compensation errors in objects orientation and excluding the possibility of damage to fragile assembly components. Bio-inspired by finray effect, a compliant gripper was designed for high-speed connector insertion, which allowed a tolerance window of up to 7.5 mm [78]. Explorations have also been tried in increasing the DoF of grippers. In addition to the close and open mechanism of the four parallel stick fingers, each finger was equipped with a servo motor, making it possible to align and grasp parts in various shapes [79].

3.2. Robotic hands

In contrast to grippers, multi-purpose robotic hands provide a much greater degree of flexibility and dexterity due to their multi-joint structure and higher DoFs. Correspondingly, they can adapt to many objects of different shapes without particular adaptation mechanism required.

A 12-DoF robotic hand was utilized to assemble modular fixture parts, where the four fingers can implement grasping, alignment, insertion and other assembly actions [80]. A multi-finger hand driven by a high-speed harmonic drive actuator could open and close at the rate of 180 degrees per 0.1 sec, which agility successfully facilitated bolt loosening detection [81]. As shown in Fig. 4(c), a four-finger hand that possessed 16 independently actuated joints, rotary encodes, and three 6-axis F/T transducers was employed to establish the multifingered manipulation control architecture, successfully performing the PiH tasks [74]. Equipped with dual-arm and two 16-DoF dexterous robotic hands consisting of four torque-controllable fingers, an advanced blind grasping strategy was proposed for a keyhole-like peg insertion [82].

Robotic hands are advantageous not only in their multiple DoFs, but also in their convenient combination with various sensors. However, very few researches on robotic assembly equipped with robotic hands can be found since there are many challenging problems to solve in the development of dexterous hands and anthropomomrphic manipulation strategies.

3.3. Task-specified EEs

Aimed at improving the assembly efficiency, a part of investigations have paid their attentions to the design of specialized EEs as per particular tasks. To adapt to PiH tasks, an adaptive EE with L-shape fingers was designed in Ref. [83], which was capable of eliminating the uncertainty of peg poses. A compliant device based on Remote Centre Compliance (RCC) mechanism was proposed in Ref. [84], that enabled the peg to be pulled out of a closely fitting hole without jamming or wedging. As for the task of screwing and unscrewing bolts from the side, a designed EE could approach the bolt in any direction through an open gear set and a width-adjustable screwing mouth (Fig. 4(d)) [75]. Softness could increase the compliance of EEs, and thus, there emerged some soft EE for assembly tasks, such as the hybrid soft material EE for reversible in-space assembly of strut components (Fig. 4(e)) [76], and the gel-based soft interface for fastener manipulation [85]. In addition, there are some special EEs that were adapted from parallel grippers to manipulate irregular-shaped flat objects and to grasp and reorientate a screw [86,87].

4. Control methods in robotic assembly

In robotic assembly, there are many variables that require precise control to maintain desired statuses, such as alignment error, position deviation, contact force, efficiency, success rate etc. In this section, we review the relevant control methods as per the basis to construct the controller. There are perception-based, constraints-based and prior knowledge-guided control methods. Although perception is still necessary in the latter two methods, introducing the constraints or taking full advantage of prior knowledge makes the control methods different.

4.1. Perception-based control

Perception is indispensable for closed loop control as it is necessary in the estimation of involved status and object information. Corresponding to the perception manner in Section 2, a variety of controllers have been established.

4.1.1. Visual servoing based control

Visual servoing based control can be classified into two schemes: position-based visual servoing (PBVS) and image-based visual servoing (IBVS). The former transforms the image measurement captured by 3D camera to errors in 3D Euclidean space and aims to reduce these errors by adjusting robots motion. In contrast, the latter adopts 2D camera and plans the motion to decrease the deviations between current and desired features in 2D image plane [88]. In the case of small-size components assembly, IBVS was adopted with a grasping control strategy and a pose alignment control strategy proposed [89]. The center point and area of target object in desired image were selected in grasping control, and the point and line features were used in alignment control. A hybrid visual servoing method combining the errors in both Cartesian space and image space were proposed in Ref. [90], where two components with 6 DoFs were assembled in 3D space. With current image features and camera pose as inputs, deep RL could be leveraged to optimize the core parameters of visual servoing controller [91].

Table 1
Typical perception types in robotic assembly.

Perception Type		Sensor	Perception objective	Assembly task	Robot	Ref.
Visual	2D vision	1 or 2 monocular cameras	Motion trajectory, object recognition, pose estimation, positioning	Cylinder block, PiH, screw assembly/disassembly, screw detection	KUKA (KR16, KR340, LBR iiwa R820), ABB IRB1200, UR5e. Dual-arm robot: Baxter, Aubo-i5	[13–17,28–32,34–38]
	3D vision	2 monocular cameras, binocular camera, structured light camera	Object recognition, assembly state estimation, pose estimation, DLO state estimation, image capture for RL	Smartphone back shell, PiH, parts recognition, cable routing	Mitsubishi robot RV1A, KUKA (LBR IIWA, KR500), UR5. Dual-arm robot: ABB YuMi	[18–23,33]
	Hybrid vision	Eye-in-hand and eye-to-hand camera, four cameras	Pose estimation	Mobile phone, PiH	KUKA KR60, ULINKHOU LR4-R5 60	[24–26]
F/T	External F/T sensor	6-axis F/T sensor, designed compliance compensator	Contact force transients, F/T for LfD/ control, CS, force tracking	PiH, electric switch, circuit breaker, folding assembly	UR10/UR5/UR3, Panda, Denso, KUKA LWR 4 + , Mitsubishi RV − 4FL; Dual-arm robot: Willow Garage PR2, ABB YuMi	[39–50]
	Proprioce-ptive	Encoder, joint torque sensor	Interaction force, contact condition, hole search	PiH	GSKK RB03A1, KUKA liwa 14; Dual-arm robot: ABB dual 7-DOF arms, Dual 8-DOF arms	[51–54]
Tactile	Traditional sensors	Artifiial skin, Capacitive tactile sensor, phototransistor-based tactile sensor	Contact classification, screw number estimation, pose estimation, alignment, diameter estimation, contour following	Human robot interaction, screw picking, boltlike fasteners, cable routing	UR5/UR3	[11,55–57]
	Vision-based sensors	Visual tactile sensor, Gelsight sensor	Pose estimation, anomaly detection, hole search, contact area and cable states estimation	Bolt insertion and tightening, PiH, cable routing	Nachi-Fujikoshi MZ04, UR5	[12,58,59]
Fused	Scanner + vision + F/T	3D scanner, depth camera, F/T sensor	Graspable points detection, success judgment, insertion force	Belt drive unit	Three arms: 1 UR3 and 2 UR5s	[65]
	Vision + force	3 cameras, a micro force sensor	visual alignment, contact force	Grasping and assembly of two components	ABB IRB1200	[69]
	Vision + F/T	Depth camera, F/T sensor	Visual and force information for RL	PiH	UR10	[67]
	Vision + F/T + tactile	2D and 3D cameras, F/T sensor, tactile sensor	Component location, pose estimation, grasp classification	Wire terminal insertion	COMAU Smart Six	[68]
	Scanner + vision	Optical scanner, camera	Object recognition	Assembly parts recognition	Unspecified	[70]
	Laser + force	Laser tracker, force sensor	Pose estimation, alignment, mating	PiH	ABB IRB7600	[71]
	Vision + laser	2 cameras, 3 laser distance sensors	Approaching, alignment	Large-scale objects	Unspecified	[72]

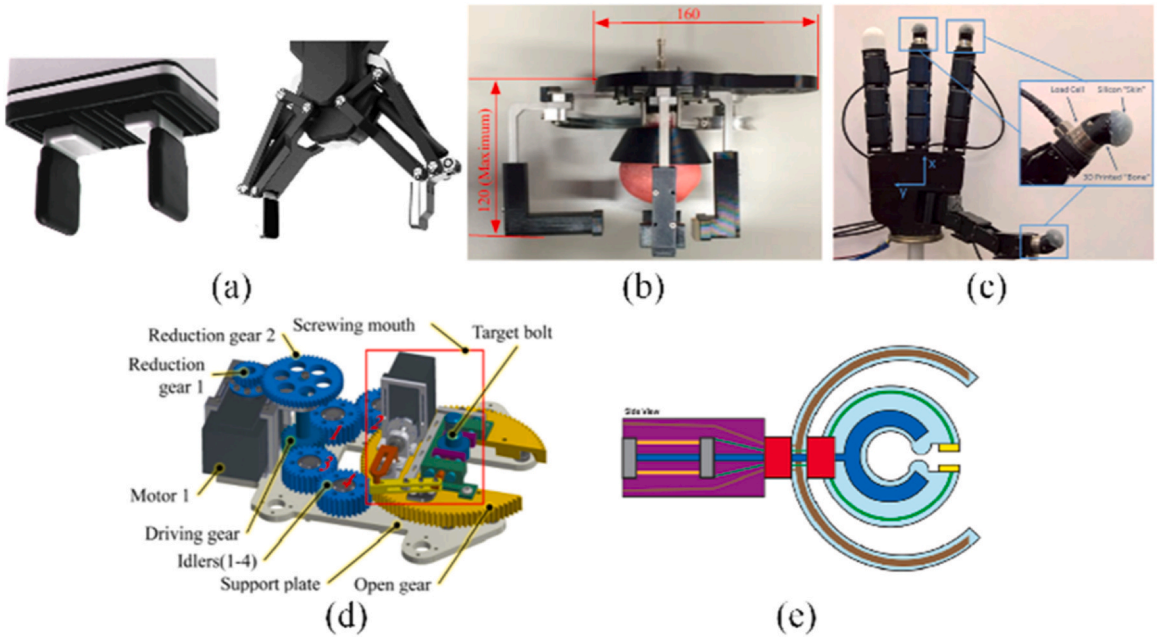


Fig. 4. Five typical EEs in robotic assembly: (a) Two-finger grippers, (b) Novel designed gripper in Ref. [73], (c) Multi-finger robotic hand in Ref. [74], (d) Task-specified EE in Ref. [75], (e) task-specified EE in Ref. [76].

4.1.2. F/T based control

F/T based control needs to be taken into consideration in contact-rich assembly as precise forces and torques not only provide necessary feedback for the motion plan of a manipulator, but also serve as control targets in compliance controllers. For force-based methods, impedance control and admittance control are two typical basis relating forces and positions. In impedance control, force and torque are usually the system output with robot motion input, and it becomes reverse for admittance control. A wrist F/T sensor is employed in these researches. With the measured F/T data, the force feedback could be used to modify the reference trajectory for safe PiH assembly [92], compute the contact point kinematics in folding assembly [49], and construct learning-based controller [93]. By establishing an admittance controller, force/torque measurement can be incorporated into deep RL to determine next actions [94] and provide compliant insertion [32]. In a more complex case of multiple PiH assembly, a novel impedance control method was proposed consisting of instantaneous center of rotation controller and variable angular velocity controller [50]. Meanwhile, force and moment produced in the interaction process can be used to control the alignment process and tackle contact state uncertainties in compliance control [41,95]. Since the system status and parameter are variant in a dynamic system, it is of great practical significance to cope with this situation. Piecewise compliance controller after the modeling of decoupling error between pose and force/moment, a machine learning-based parameter optimization, and a transferable force controller based on prescribed performance control were proved valid solutions [42,48,96].

4.1.3. Hybrid perception based control

In general, both visual servoing and F/T based control are expected in a complete assembly system to handle both non-contact and contact scenes. To imitate a manual-skillful action in PiH task, a hybrid switching force/position control scheme was developed, and both visual and force measurement were implemented [97]. A visual servoing scheme consisting of PBVS and IBVS and a compliance force control using F/T sensory data were combined together, realizing the pose alignment and placement [98].

4.2. Constraints-based control

During an assembly process, the contact and interaction between robot, EE, objects to assemble and environment are inevitable. Hence, constraints widely exist in a variety of tasks, e.g., the geometric constraint and force/torque constraint, which can also be utilized to ease the assembly strategy.

4.2.1. Attractive region in environment

Geometric constraint from environment is the most common form. A typical constraint concept is the attractive region in environment (ARIE), which is defined as a region formed by the environmental constraint in the configuration space of the robotic system, e.g., a hole in a PiH task. As it is able to achieve high-precision manipulation in low-precision systems, ARIE can be applied to the grasping, localization, and assembly [99]. As for PiH assembly, it has proved effective by constructing the region defined by geometric constraints between a peg and a hole [41,100]. The constraint region could be designed for motion planning and used for uncertainty elimination induced by initial positioning error [100]. It was also found that ARIE could accommodate the transition between different CS and model the repositioning of peg with respect to hole [41]. A similar concept is the attraction basin in hole vicinity, which was defined as the area surrounding the center of a hole [101]. Its geometrical features, such as chamfers, were found essential in determining the size and shape of the attractor basin.

4.2.2. Other constraints

The structural constraints could be described in other forms. For example, a constraint graph structure was proposed in Ref. [102], where each vertex represented a block and a pair of directed edges corresponds to the contact. With it, a dual-robot system implemented the assembly of 48 blocks, forming an interlocking cube-like structure. As for shallow-depth insertion, the shallow hole was postulated as fixed and stiff constraints [103]. The frictional contacts and contact normals decreased the motion DoF and simplified the modeling. To respond to external force constraints, the level of robot compliance was regulated in constrained directions, which coped with uncertainties in vision and dynamics [104]. Force feedback could also be used to plan active and

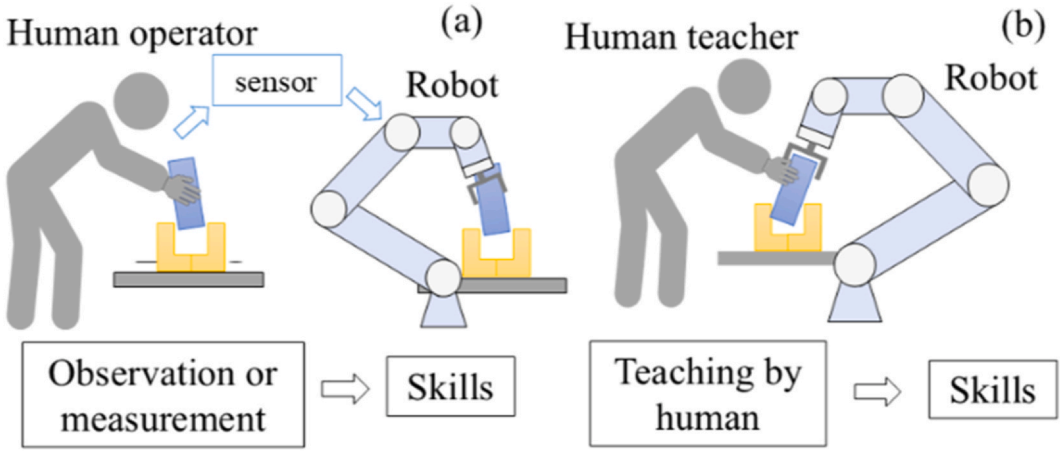


Fig. 5. Two methods to obtain demonstration data: (a) Observation or measurement, (b) Teaching by human.

passive compliant motion, ensuring the manipulator state to always stay inside the constraint region [105]. In addition, incomplete constraint could also be leveraged, such as the imperfect borders of holes and cascaded shafts, which could be solved by modifying the CS transition sequence and contact models [106].

4.3. Prior knowledge-guided control

In addition to the constraints, how to take advantage of prior knowledge in advance of task execution deserves attentions since it facilitates the robots to improve the assembly accuracy and decrease execution time. These knowledge could be the geometric template of objects to assemble, CS, skill experience from human demonstration, and other transferable information from different tasks.

4.3.1. CAD model guided control

Among various prior knowledge, CAD is frequently used due to its accessibility in most assembly scenarios. It is able to provide the part dimensions, shape details and geometric features as well as the mating descriptions between different parts. Component poses could be extracted from 3D CAD models and calibrated via visual perception, based on which the component localization and a flexible robotic assembly system was developed [107]. Offline data from CAD and online data from the vision system were helpful in compensating the uncertainties and enabling modular robot control architecture [108]. The assembly model and semantic description could be extracted from the CAD data. Furthermore, the assembly constraints, robot motion constraints and their relations could form a knowledge ontology, which assisted in grounding from CAD-level specification to suitable skills [109]. Besides, CAD model was able to help the robot produce grasping strategies through grasping effect evaluation by CAD models of the grasping finger [110], or through the design file containing the jigs and gripper at different assembly stages like grasping and placing [111].

4.3.2. Other prior knowledge

In addition, a lot of investigations tried to utilize other prior knowledge. Assembly direction, end-condition, rough location and insertion area of the male and female components, and the primitive knowledge base could be exploited to construct a multimodal controller in PiH task [112]. An observer algorithm that monitored the assembly progress and provided state information for high-level controllers was proposed with sequential Monte Carlo algorithm combined [113]. The observation of assembly processes helped error handling, action adaptation and robot controlling. By creating a comprehensive library of skills, the reuse and incremental refinement of assembly skills were explored as prototype-based inheritance [114]. Instead of instantiating a new class, cloning an existing object implemented the fine-grained inheritance mechanism. Mesh data representing the objects geometry, rough estimate of the hole pose, and a desired goal state were inputted into a state estimation framework to reuse the samples in motion generation step and establish a robust impedance controller [115].

Compared with off-the-shelf knowledge, low-cost exploration is an active manner to produce prior information for future assembly tasks. A fuzzy logic-driven exploration strategy based on Cartesian compliance controller was proposed to generate knowledge required by subsequent model-based RL method [116]. It could not only relieve the training process from suffering of data size requirement, but also improve robustness of the Cartesian compliance controller.

5. Learning methods of assembly skills

Learning from human demonstration or the interaction between an agent and the environment have received intense attention in recent years, which could bypass the difficult modeling process of underlying mechanism. It not only saves a lot in cost, but often brings with satisfying results. Besides, skills transfer from existing models to new tasks is preferred in many tasks.

5.1. Learning from demonstration

Learning from demonstration (LfD), or called learning by demonstration (LbD), refers to the acquisition of assembly skills from human demonstration by imitating a human operator. There are mainly two methods to achieve the demonstration data: Observation or measurement, and teaching by human, as shown in Fig. 5. For the first method, no interaction is required between the human operator and the robot. The assembly process is recorded, i.e., images or videos are obtained via 2D or 3D camera, or measured by external sensors, e.g., force/torque sensors, haptic gloves, leap motion, and reflective markers attached to human body. As for the other method, a human teacher needs to guide the robotic arm and EE to finish the assembly task via a teaching device, during which process the pose of EE, joint angle of robot and produced forces/torques are obtained and stored.

The learning process is in essence an abstraction of the assembly strategy rather than a replication of the demonstration data. According to the abstraction and representation methods of human demonstration, four prevalent approaches can be found in existing investigations: Gaussian mixture model (GMM) and Gaussian mixture regression (GMR), Hidden Markov model (HMM), Dynamic motion/movement primitive (DMP), and deep learning (DL).

5.1.1. GMM/GMR

As a typical ML algorithm, GMM is used to calculate the probability of data point falling into a particular cluster. It can be interpreted as the mixture of a combination of multiple Gaussian components, which

constitute a multi-modal distribution. Expectation-maximization (EM) is usually adopted for GMM training. It is characterized with three critical parameters: mean, variance and weight.

In robotic assembly, GMM and GMR are often used simultaneously. GMM is first used to estimate the distribution parameters of each Gaussian component, and then GMR is applied to approximate the mapping relationship between the known variables and unknown quantities or status. In LfD, after learning the statistic parameters via GMM, GMR is most used to generate the trajectory of EE of a robot, including the 3D position and spatial angles at each time step. The taught demonstration was transformed into aligned pose sequence at first, and then the end trajectory was computed with GMM/GMR, which controlled the robot motion after inverse kinematics (IK) [117]. The demonstration data obtained via an optical motion capture platform was used to learn the GMM policy, i.e., robot motion trajectory, after outlier identification and trajectory segmentation in 3 C assembly [118]. Using a teaching device, successful demonstration data was obtained and GMM/GMR was leveraged to create the reference trajectories after labeling in advance [119]. The low-level skills generated with GMM/GMR were proved effective in new environmental conditions [120].

Besides, the GMM/GMR method could facilitate to estimate feasible force field [121], obtain the experimental force-pose-deviation curve [122], fit the dynamical system and infer the coupled state [123], and process image information for robot grasping [124]. Furthermore, GMM/GMR was able to help construct an admittance controller for LfD [125,126].

5.1.2. HMM

HMM is adopted to model the relationship between observations and their dependent states, satisfying the Markov decision process (MDP). Its parameters are often estimated via the maximum likelihood estimation and Baum-Welch algorithm. In robotic assembly, HMM is widely used in various applications, such as the recognition of dynamic gestures for robotic manipulation in tele-operation [127], the shortest assebmly path planning under MDP frame [128], and the parameter identification for variable admittance control [129]. In the application of RL to robotic assembly, HMM was utilized to learn the assembly policy [130] and reward values from very few demonstrations [131].

After the implementation of a limited number of human demonstrations and recording the robot EE pose, HMM was applied to select the nominal reference trajectory, thus assembling a gear into its square-section shaft [132]. A task-parameterized hidden semi-Markov model (TP-HSMM) was proposed to achieve weighted contribution of tele-operated demonstrations [133]. By estimating the joint distribution of the tactile sensor input and action outputs observed in a demonstration, HMM approach and modified GMR formed a LfD framework and succeeded in an edge-following and manipulation task involving different object geometries [134].

5.1.3. DMP

Dynamic motion/movement primitive (DMP) represents an elegant mathematical formulation of the dynamic motor primitives and is well suited to generate motor commands for robots [135]. To plan a robot motion, the trajectory can be defined by a second order linear dynamic system with an additional nonlinear term, containing the free parameters to adapt generated movement to demonstrated trajectory.

In the assembly tasks on the Cranfield benchmark, the interaction between manipulated objects was captured by a vision system, and applied to compute the desired position from DMPs in Cartesian space [136]. Robotic wire pinning for wire harness assembly was realized by the primitives and vision-guided control learned from a human technician and a collaborative robot [137]. Both position and orientation data in demonstration phase were sampled to generate a DMP library and activate a DMP's nonlinear forcing function during execution. In a PiH task, GMM/GMR was used to extract the optimal trajectory encapsulating sufficient human features and DMPs were applied to learn the trajectory generalization [138]. Considering the defects of traditional DMPs in dealing with assembly jamming, a reversible DMP method was proposed in Ref. [139], which encoded the demonstration trajectory, time, and force/torque profiles in a phase transform system and then reproduced them with an admittance controller.

5.1.4. DL

The DL technique has progressed enormously these years. Among various functions, classification, regression and control constitute the three main subfields, which are required in most robotic assembly applications. Classification helps robot recognize different assembly objects and segment skills or trajectories, with CNN and its multifarious variants, e.g., YOLO and ResNet, as the representative. Regression can be used to model the mapping relationship between different variables and status. For instance, the most widely used recurrent neural network (RNN), e.g., LSTM (Long short-term memory), is suitable in predicting assembly sequence related to time series. In addition, the typical control model, deep RL (DRL), is capable of replacing conventional control frameworks and achieves satisfying robustness and accuracy.

To improve the accuracy and generalization of DMPs, a CNN-bsed segmentation method of the insertion location was proposed to segment the task into several movement primitives during different phases including pickup, approach, insertion and retraction [140]. With the tuple data $[x, \dot{x}, f^d]$ as the input of a LSTM model, it predicted a sequence of elements f^d for N steps, serving as reference setpoint for the force regulating PD controller [141]. By observing human demonstrations of the work process and content, DL-based operation sequence generation and target pose of assembly objects were successfully realized in Ref. [142]. CNN, Resnet, and YOLO models were used constituting a hybrid DL network. In the learning of robotic insertion from human demonstration, a latent skill-guided RL approach was leveraged and a new programming by demonstration (PbD) based learning framework was proposed [143]. How to transfer human skills to a robot within a safe region was also explored.

5.2. Learning from interaction

In LfD, the assembly skills are usually learned by observation or manual teaching, which doesn't focus on the frequent interaction between robot and the environment. In this section, we mainly focus on how to learn the skills from an implicit process where the system makes decisions according to the acquired experience of interaction with environment. Although there are other interactions, such as the human robot interaction (HMI) in robotic assembly [144], we limit ourselves to the complete automatic robotic assembly and the interaction occurs between the robot and its environment. The most typical case is RL that learns experience from the interaction between an agent and the environment via state estimation, reward calculation and adopted actions in simulation or real experiment. Current RL algorithms that have been applied to robotic assembly can be classified into model-based type and model-free type, determined by whether an interaction model is adopted [145]. Further, model-free RL includes value-based RL (VBRL), policy-based RL (PBRL) and Actor-critic RL which combines the advantages of VBRL and PBRL.

5.2.1. Model-based RL

If the exact transition dynamics of the environment is given in an assembly task, model-based RL is preferred since it is able to decrease the number of iterations to a large extent and saves the computation cost. The dynamic model can be used to derive the system states and calculate the reward before next action is made. Among various MBRLs, the guided policy search (GPS) method utilizes refitted local linear models to optimize trajectory distributions, which not only speeds up the learning process, but does not rely on a global model [146]. In the assembly tasks of U shape, gear and PiH, GPS was designed accounting

for two components: trajectory optimization component learned the time dependent models, and the supervised learning model learned a state-dependent but time independent NN policy [147]. Iterative Linear-Quadratic-Gaussian (iLQG) was applied in Ref. [148] considering it is sample efficient and convenient to solve quickly with second-order methods, which was combined with space force control and hybrid motion/force controller for stable behaviors. A robotic industrial insertion learning method based on model-based meta-RL was proposed to improve the learning efficiency in order to insert an E-model electrical connector into its socket [149]. In Ref. [116], a MBRL agent, a Cartesian compliance controller and a knowledge-guided exploration strategy constituted an assembly framework. Therein, the MBRL agent was established with NN to take full advantage of model-based data generation techniques and model-free policy optimization.

5.2.2. Value-based RL

As for value-based RL, explicit policy is not required and the value function of each state is calculated using the temporal difference. Q-learning is one of the popular VBRLs that learn the value of an action to find out the optimal action-selection policy. Q-value denotes the value of taking a specific action within a particular state for given finite MDP. To solve the uncertainty in complex assembly process, the CS was described by force/torque as a dynamic description of the assembly process based on data [150]. The skill of pose adjustment was mastered by the deep Q-learning model.

Under the premise of Q-learning rule, a DL network can be established to extract the low-level features and approximate the action-value function, which is called deep Q network (DQN). Considering that the influence of visual information on the assembly state was ignored by most works, [151] proposed a skill learning method based on DQN using both visual perspectives and force sensing to learn an assembly policy. The reward system was designed with an image template matching for assembly state while the observations were described by force/torque information and the pose of EE. In addition to the trajectory planning via selecting the stiffness matrix [152], DQN could be leveraged to construct an integrated framewor comprising state judgment DQN and parameter optimization DQN, which was then used to train a variable impedance controller (VIC) [153]. A double DQN (DDQN) RL was applied by Ref. [154] and the observation, action and reward was generally designed for dynamic flexible job shop scheduling. Multi-agent RL system was constructed to overcome the lack of flexibility and autonomy of single-agent centralized model. An advanced version of DQN was proposed, called twin-smoothed multi-pass DQN, which was capable of relieving the problem of Q-value over-estimation [155].

5.2.3. Policy-based RL

In PBRL, the policy is explicitly constructed with each state mapped to the corresponding best action and the parameterized function kept in memory. Estimation of the value function is no longer required and a continuous spectrum of actions was allowed. Goal-conditioned action primitives that were decoupled with a meta policy were applied and an inverse dynamics model was used to facilitate interpretable goal state and reachable trajectory [156]. Based on a stable VIC controller as the policy parameterization, a novel policy search algorithm was introduced by Ref. [157] to inherently guarantee the stability, which was inspired by Cross-Entropy Method. As PBRL suffers from slow convergence and high variance, it is seldom employed directly in robotic assembly.

5.2.4. Actor-critic RL

Actor-critic RL (ACRL) is a combined version that tries to embrace the advantages of VBRL and PBRL, which learns the policy and value function simultaneously. Therein, the actor generates policies and the value function is used by the critic to evaluate the policy. How to include the expert behavior in an ACRL setting with simple reward function and continuous state-action space was investigated to establish an admittance controller [158]. Success in a PiH task validated the combination of RL, behavior cloning and admittance control. Soft actor-critic (SAC) was an algorithm that optimizes a stochastic policy in the off-policy way, which seeks the maximization of both lifetime rewards and policy entropy. Because of its excellent performance, SAC was widely used in robotic assembly, such as multimodal policy construction of adaptive compliance control [159], image-based trajectory generator that determined the robot movement direction [160], and combined framework through hybrid trajectory and force learning [161]. It was modified as a Actor-Average-Critic to achieve better stability and sample-efficiency performance [162]. A data-efficient Hierarchical RL (HRL) was proposed containing two augmented MDPs, in order to learn both higher level and lower level policy from the same samples [130]. It was successfully applied to dual and triple PiH assembly.

Deep deterministic policy gradient (DDPG) algorithm belongs to another popular ACRL type which has also been extensively applied in robotic assembly. DDPG learns the deterministic policies and expands them into a continuous action space through the actor-critic architecture [163]. In Ref. [164], an actor-critic approach was adopted based on the deterministic policy gradient algorithm, which included four networks, specifically, both critic and actor network contained target-net and eval-net. To tackle different assembly problems, a variety of variants of DDPG have been developed. One of these variants, Ape-X DDPG, was found well suited to tackle complex assembly tasks benefited from its distributed architecture and much higher training efficiency [165]. With recurrency, it was further extended to the Recurrent Distributed DDPG (RD2) which made two structural improvements on prioritized experience replay [166]. When applied to two fundamental high-precision assembly tasks, lap-joint and PiH, it outperformed the Ape-X DDPG and PPO with LSTM. A model-driven DDPG algorithm was proposed to learn high-level assembly policy, where a traditional force control strategy was adopted rather than learning from zero basis and instead of complex reward function, a fuzzy reward system was utilized [167]. It not only decreased the number of training experiments and avoided risky actions, but also improved learning efficiency. A fuzzy logic-driven variable time-scale prediction-based RL was developed for assembly action control to improve the efficiency, which was found effective in decreasing the assembly time when combined with DQN and DDPG [168].

Another variant of ACRL is proximal policy optimization (PPO) algorithm. Two restrictions were added to the objective function of PPO to avoid unstable policy update caused by the step size and policy ratio. In the case of contact-rich assembly characterized by multiple solid-body contacts, PPO-based RL was used to assemble the automotive air extractor part, in which a fully connected dense network was designed acting as the policy [169]. In Ref. [170], while a DQN model was applied as a grasp-net network to learn the grasping skill operations, the center point coordinates and rotation angle obtained through principle component analysis (PCA) were inputted into a PPO network as prior knowledge and the optimal actions could be outputted.

The characteristics of various RL methods in this section is summarized in Table 2.

5.3. Skill transfer

When an assembly skill is acquired by a robot either from a simulation environment or a real-world experiment, a problem need be taken into consideration: how to transfer the learnt sill from a source domain to a target domain while the loss of accuracy is minimized? In this section, the issue on skill transfer is reviewed and two situations are discussed, including simulation to reality (Sim2Real) and adaptation to new tasks.

Table 2
Characteristics of the RL methods in Section 6.2.

Method	Ref.	Algorithm	State space	Action space	Assembly task	S/R[a]	Success rate
Model-based	[147]	GPS, iLQG	Joint angle & velocity, EF position & velocity	Position & velocity	U shape, gear, PiH	S & R	0.9 (S); 29/30 (R)
	[148]	ILQG	Joint angle & velocity, EE position & velocity	Joint F/T	PiH, gear wheel and hole on shaft etc.	R	16/20 for 4 tasks (R); 23/30 for generalization
	[149]	Meta learning	EE pose	3 translations	PiH	S & R	14/15 (R)
	[116]	SAC	EE pose & Velocity, F/T, goal pose of the hole	EE velocity	PiH	S	98%, 91% (S)
Value- based	[150]	Q Learning	F/T	6 rotations	Low-voltage apparatus	S & R	84.2%, 80% (S); 80.8% (R)
	[151]	Q Learning	EE pose, F/T	6 rotations	Circuit breaker	S & R	97%, 90% (S); 91.4% (R)
	[152]	DQN	Tip position, F/T	Stiffness matrix	PiH, gear insertion	R	100% when position error $<$ 3 mm (R)
	[153]	DQN	F/T, z and its offset	Impedance parameters	PiH	R	96% (R)
	[154]	DQN	Position and other 5 types of variables	Accept, Wait, Decline, Continue	Job shop scheduling	S	Not provided
	[155]	DQN	Pose between peg and hole, peg vel & wrench	Translation, rotation, insertion and action parameter	PiH	S & R	68.6%−94.0% (S); 86.7%−100% (R)
Policy-based	[156]	GAN-augmented meta policy	EE pose & force, door frame pose, door handle position	Approach, grasp, move2target	Relocation, hammer-use, door-opening	S & R	28%−92% (S); 8%−92% (R)
	[157]	Cross Entropy method	Position & Velocity	F/T	PiH	S & R	No quantitative data provided
AC	[158]	Cycle of learning	Pose, F/T, controller target	Pose change	PiH	S & R	100% and 62% (S); 8% and 82% (R)
	[159]	SAC	EE pose & Velocity, insertion force, previous action, F/T features	Pose, controller parameters	PiH	S & R	100% (R)
	[160]	SAC	Position, velocity, F/T	Position & Velocity & acceleration	PiH	S & R	Over 95%
	[161]	SAC	EE pose, F/T	Pose, controller parameters	L, condenser, HDMI insertion	S & R	Not provided
	[162]	AAC	Peg pose, F/T	Joint angle & Velocity, force, controller parameters	PiH	S	Not provided
	[130]	DPG	Peg pose, F/T	Pose	PiH	S & R	Not provided
	[164]	DDPG	Joint angles & Torques, EE pose, F/T	Joint angles	PiH	R	94% (R)
	[165]	DDPG	Pose, F/T	Linear and angular velocity	Timber joints	S & R	100% Single-lap task (R)
	[166]	DDPG	EE F/T	Linear and angular velocity	Lap-joint, PiH	S	100% without offset and noise (S)
	[167]	DDPG	Peg pose, F/T	Translation and rotation	PiH	S & R	100% (R)
	[168]	DDPG	Peg pose, F/T	Impedance parameters	PiH	R	100% (R)
	[169]	PPO	TCP pose, F/T	Translational and rotational change of TCP	Automotive assembly	S & R	100% (S); 100%, 85% (R)
	[170]	PPO, Q Learning	Pose of parts and EE	Center point and rotation angle	PiH	S & R	90% (S)(R); 86.7%, 73.3% under 3 mm tolerance (S)(R)

[a] S: simulation, R: Real-world experiment.

5.3.1. Sim2Real transfer

As most RL algorithms and plenty of controllers rely on simulation to finish the training of models, the reality gap does exist when these models are transferred from simulation to reality (Sim2Real) and it usually results in the performance degradation due to their discrepancies. The sources of deviations include the inaccuracy of simulated models, uncertainty in sensors and multifarious environmental factors. Domain randomization is employed by most investigations to reduce this reality gap, which takes the variance of real environment into consideration from the statistical perspective. Two types of randomization were used in Ref. [171] including observation randomization and dynamic randomization. Uncertainty of the target and initial state, and the inaccuracy of model and interacting strategy were considered, with Gaussian distribution and uniform distribution of noises added to the initial joint angles and actions. Furthermore, a Sim2Real controller was designed to decrease the position error between simulation and reality. Domain randomization was also able to simulate the noise of real environment, e.g., background color, camera parameters, lighting condition, and robot dynamic parameters [172]. An impedance-based controller could map the desired EE pose into real joint torques to aid the RL agent's implicit learning and couple the policy during simulation and actual experiments.

Meanwhile, improving the simulation fidelity was a potential way to narrow the reality gap and the prevalent methods include contact reduction approaches, real2sim frameworks, system identification through robotic interaction and so on. The essence of contact reduction approaches was to limit the number of contact points which slowed down simulation and might led to numerical instability, and the simulation accuracy was successfully improved for a double pin insertion task [173]. In real2sim frameworks, differential physical engines are often applied allowing the computation of gradients with respect to physical simulation parameters. Therein, the gradient enables the optimization of system parameters. A real2sim2real framework was proposed in Ref. [174] to address the sim2real gap by separating the whole process into a real2sim training phase and a sim2real inference phase. An inverse idea was proposed by Ref. [175] to apply real2sim in the test phase. In the inference part, the real physical robot took it as working in simulation, which was generated by a generative adversarial network (GAN) model. It transformed the real scene into an image similar to the simulator and then reduced the discrepancy between simulation and reality. A novel real2sim approach to unsupervised domain adaptation was constructed for object pose estimations, which preserved the structural and semantic information, and object pose during transfer [176]. In addition, a neural surface reconstruction method was leveraged to reconstruct high-quality meshes from real point clouds to facilitate fast simulation, and a real2sim2real pipeline was validated in better sample efficiency [177].

5.3.2. Adaptation to new tasks

Adaptation of acquired skills from existing tasks to new unseen task or a similar task in different environment is also faced by a mass of robotic applications. One possible method is transfer learning (TF), which pursues to improve the learning accuracy in target predictive function using the knowledge in source domain. Due to the domain difference, direct reuse of learned policy most probably leads to failure and thus, various measures were taken. A novel transfer RL method, called TRANS-AM, was proposed based on aggregating dynamics models [178]. It learnt to aggregate source dynamics models adaptively in MBRL loop to better fit the state-transition dynamics of target environment and execute optimal actions. A policy fusion transfer algorithm was designed in Ref. [179] by fusing the policies between the source and target domain MDPs. According to the data obtained from the interaction between the fused assembly policies and the environment, the target domain assembly model was trained to fulfill the policy transfer. A feature-selected adaptation transfer was realized based on domain adaptation [180]. Specifically, the source domain data that had small maximum mean discrepancy to the target domain was extracted to pretrain the target domain policy. Besides, the data that was more favorable to the target domain were selected to update the network. Its effect was validated in a series of PiH tasks.

6. Performance evaluation

As for a robotic assembly strategy, how to evaluate its performance and estimate the obtained result is of high importance. In this section, we summarize the generic evaluation metrics that are widely adopted in robotic assembly. The multifarious metrics employed in existing works are reviewed in five categories: accuracy-related, time-related, error-related, force-related, and reliability-related metrics. There are also a few other metrics, such as the learning related metrics and refer to Ref. [7] for more details. Particularly, in PiH task *clearance* is a key parameter, given in the form of ratio or deviation between the diameter of the peg and the hole. In precise assembly, it may be as small as 0.01 or 0.03 mm [105,142]. A millimeter and sub-millimeter clearance is used in most assembly scenes [41,52,112,122,129]. It is often used as an experimental configuration parameter rather than an achieved result.

6.1. Accuracy-related metrics

Success rate (SR) is widely adopted as the criteria to assess the performance in most assembly task. Specifically, $SR = N_{successful}/N_{total}$ where $N_{successful}$ and N_{total} are respectively the number of successful trails and total trials either from simulation or real-world experiment. In many existing works, SR can be as high as 100% [20,90,122,178]. It is also adopted by a few grasping and pose alignment tasks [89]. Meanwhile, the number of total trials, N_{total}, is given to increase the fidelity of SR. In general, a larger value of N_{total} is expected although it distinctively different in existing works, such as 5 trials under each situation [94,148,149], and 600 total trials [54].

In perception tasks, *recognition accuracy* or *classification accuracy* is often used to evaluate the recognition performance, which significantly affects the final assembly result. Its computation is similar to SR, i.e., the portion of correctly recognized trials over total trials. Such cases include the recognition accuracy of pins [15], the hole finding accuracy [29], discrimination accuracy of different actions [40], classification accuracy of CS [44], and differentiating accuracy between normal and faulty insertions [47].

6.2. Time-related metrics

Completion time (CT) refers to the time required to finish the target assembly task. As fast assembly is expected, a smaller value of CT indicates higher efficiency. CT varies with respect to the complexity and hardware configuration. It is adopted by many PiH tasks [29,50,52,74,173]. In visual perception, the image processing time also deserves attentions [14]. Identification time for threaded holes and bolts is an indicator for the recognition algorithm [27]. In the comparison with other strategies, reduction time is used to indicate the improvement degree, such as the reduction of assembly time by 15% in Ref. [39]. In execution phase, the execution time or computational time is also cared about [11,41,46]. As for skill transfer, *settling time* was defined as the smallest time after which the tracking error remained inside a tolerance band, and it represented the response speed of a controller [48]. *Training time* or *testing time* is an indicator for some learning algorithms [44].

6.3. Error-related metrics

Positioning error or *orientation error* is important for the pose estimation of an objects [14,19,37]. They respectively denote the difference of translational values along and the orientation angles about *x*-,

y-, and *z*-axes between desired and estimated variables. Sometimes, its *maximal value*, *mean absolute value* and *standard deviation* is used to conclude a final indicator across multiple trials or dimensions [17,32,37,98]. It is also used to characterize the uncertainty degree and further verify the robustness and reliability of assembly skills [149,159].

6.4. Force-related metrics

As robotic assembly is a contact-rich scenario, *contact force* is inevitable and needs to be estimated frequently [43,49,93]. In PiH task, the smaller the contact force is, the better the assembly skill performs since it means less abrasion and higher success rate in most cases. It can be measured with F/T sensors. *Average force* and *maximal force* are assessed in some PiH tasks [116]. In developed EEs, *grasping force* is measured as a characteristic parameter [86], and a larger value is preferred.

6.5. Reliability-related metrics

In addition to above metrics, how to judge if an assembly skill is reliable also needs to be taken into consideration, especially in practical applications. Usually, the assembly process will be repeated for many times and statistical measures are calculated over the involved important indicators. One way is to plot the *position histogram*, *posture histogram*, and *repeatability error* after repeated measurement [25,43]. *Analysis of variance* (ANOVA) can be leveraged to compare the variation of multiple groups of data. In Ref. [74], the ANOVA test was used to determine the difference in sample means of two completion time datasets (Continuous data), and the Kolmogorov-Conover algorithm was applied to check the statistical difference between two sets of pass-fail data at a 95% confidence level in the case of discontinuous data. Another manner to evaluate the reliability of assembly result is repeated evaluation of certain metrics under different uncertainties. These uncertainties include the change of clearance fit, different noise levels, random initial states, and variant environmental perturbations [116,156,172,181].

7. Conclusion and future directions

In this paper, the key technologies in robotic assembly are summarized. On one hand, the necessary key hardware components are described, such as the core sensors and their design, principle, and arrangement. As for an assembly task, there are usually multiple options to establish the system, accompanied by respective merits and shortcomings. On the other hand, the algorithms, models and other state-of-the-art techniques are discussed. Through existing works on robotic assembly, it can be found that the majority of researches focused on three types of assembly tasks:

(1) *PiH task*. As revealed in Tables 1 and 2, PiH has received intense attentions from various aspects. The seemingly simple task becomes rather challenging when small clearance between the peg and hole is encountered, specifically 0.02 mm to 0.2 mm in precise assembly [182]. Although single perception technique has been proved valid in PiH task, there still exist lots of assumptions and limitations hindering its application in unstructured environment, and fused perception is recommended considering high precision and compliance are thought as its two main challenges. In most PiH tasks, the peg geometry appears relatively regular and symmetrical, and hence, no special requirements are posed to the EE. Gripper, robotic hand and specified EE are feasible in implementing the alignment and insertion actions. In addition to various control methods, ARIE is especially effective in PiH assembly. To train a RL model, EE/peg pose, F/T and velocity often constitute the state space, and joint angles, positions, and other action parameters are used mostly to form the action space.

(2) *Bolt screw assembly*. Albeit it appears very similar to PiH task, the bolt screw assembly has its unique characteristics. There are many types of bolts, e.g., ordinary bolts, hexagon fit bolts, studs, anchor bolts, and carriage bolts, and their sizes also vary, which brings more difficulty in the assembly. Compared with PiH, the assembly process additionally includes selection and taking of the bolt of correct types and size in perception procedure, bolt/nut mating, and bolt tightening [8]. Especially, the bolt tightening control not only requires the use of a tightening tool, but also precise fused perception and accurate robotic control unit. Robotic gripper is no longer suitable in this situation, and robotic hand or task-specified EE is possible to implement the complicated procedures. In coordinated bolt tightening control technology, the fault conditions such as thread, stagnation and slippage need to be taken into consideration.

(3) *Cable routing and other assembly tasks*. Currently, investigations on robotic cable routing are relatively rare due to its high flexibility, non-linearity and deformable property. However, it is inevitable in the assembly of electronic products. The perception techniques, such as 3D vision guided, tactile enabled, or 'vision + F/T + tactile' fused perception facilitated routing, have been explored, and the main perception information has been listed in Table 1. While the influence of grasping position on the cable deformation can be modeled and analyzed, path planning deserves particular attentions and LfD may be an effective solution [183]. Different from the above fundamental tasks, some researches have turned to the assembly of more complex products, e.g., smartphone back shell, electric switch, belt drive unit, low-voltage apparatus, and circuit breaker, as shown in Tables 1 and 2. Actually, most such tasks can be divided into multiple basic assembly scenes like PiH, bolt/screw assembly and cable routing. Thus, the involved techniques we introduce in this review still work in these cases.

Although great progress has been achieved in past years, there are still several challenging issues to explore, which reveals the potential directions in robotic assembly.

(1) *Precise perception*. As analyzed in Section 2, the prevalent sensors selected in assembly system is still cameras and F/T sensors, which suffer from indirect sensing information of contact points and rather high requirement for processing algorithms. Currently, numerous novel sensing techniques have surged in other fields, such as tactile sensors, laser sensors, proximity sensors, ultrasonic transducers, and radio-frequency sensors. Fused perception using multiple types of sensors is a very potential direction to realize precise perception and increase the system's robustness, as a few pioneer works have shown its great value [69–72]. However, most existing fused perceptions focus on the direct integration of multiple sensors, and seldom works can be found on the multimodal perception mechanism, which still stays challenging. As multimodal perception could greatly enhance the measurement accuracy, it deserves further in-depth investigation. Besides, based on our analysis in Section 2.3.3, a lot of advanced tactile sensing techniques have not been applied to robotic assembly but exhibit immense potential and hence, it can be paid more attentions as an alternative solution of F/T sensors, which provides direct and flexible perception of the property and states of the objects to assemble [184]. Thus, it not only enables reliable in-hand manipulation, but make it possible to tackle more complex situations [185]. A potential direction is to convert the tactile data into images and then realize multimodal fusion by feeding these images and visual data into the same DL model, since this transformation has been verified effective [186].

(2) *Dexterous hand*. In existing researches on automatic assembly, grippers are widely adopted as the EE due to their relative low cost and reliability. However, it can be found through existing works that the grasping states and in-hand manipulation are seldom involved. In most works, several premises are assumed in advance as that the initial pose of objects to assemble and the grasping point is known. Meanwhile, it is taken as granted that the execution of grasping, movement and releasing actions are successful. Although the in-hand manipulation is

tried to handle uncertain initial status with tactile sensing, its manipulation objective is relatively limited. Therefore, in order to implement the assembly of various objects of different shapes, material and stiffness, dexterous hand should be taken into consideration mimicking human operations [185]. Meanwhile, the development of robotic hand is expected in the improvement of more DoFs for higher dexterity, embedded sensing and corresponding control models. Bio-mimetic design may be potential to assist more precise operations, such as the role of human fingernail in the insertion of a micro plug.

(3) *Error Recovery*. When an assembly error is produced in manual operation, the human operator can easily undo the process and restart the assembly until the desired assembly state is reached. This error recovery procedure is necessary in practical applications especially for complex assembly occasions. However, most existing works excessively focus on the success rate at one time. Two processes are required including the error detection process and error recovery process successively. In contrast to the differentiation of a faulty or normal assembly, which type of errors the system is being faced with more matters the subsequent actions. Knowledge-based and graph-based approaches are two potential solutions, which facilitate the system to recover from errors by predefined actions and graph search respectively. For example, an automatic error classification method was proposed to discriminate four error states when a battery was inserted into a cell phone [187]. For different error types, a corresponding error recovery strategy was predefined. Four blind search methods were found effective for error recovery when the insertion of five types of commercial electronic connectors met with uncertain position errors [188]. Reversible execution could transition the system from faulty states to initial normal states in most assembly tasks [189]. How to achieve high accuracy of error detection, find out the error source and recover the system to a known states deserves more explorations in the future. It is of high importance to increase the reliability of an assembly system.

(4) *Collaborative Robot*. Along with the development of humanoid robots, the necessity of collaborative robot in robotic assembly deserves intense attentions. It seems that the adoption of single manipulator has satisfied the requirement of most assembly scenarios. However, there exists a lot of limitations for single-arm robot. Firstly, the operational space is limited leading to more constraints on the selection of assembly objectives, positioning accuracy of robot itself, and path planning algorithms. Moreover, the singularity problem should also be taken into consideration. Secondly, more prerequisites are necessary to construct a relatively structured environment. For instance, the object to assemble is expected to locate at a known position in a predefined pose. Even if the in-hand manipulation capability was equipped by tactile grippers allowing unfixed initial pose [57], several preconditions were still be predefined, e.g., the hole to inset was fixed, the start and target position were given, and the bigheads lay on a rail. Thirdly, it remains difficult for a single-arm robot to learn how to pick and use a tool for assembly. These difficulties can be eased by using a collaborative robot. While one type of the collaborative robots is the cooperation of multiple individual robots [190,191], the dual-arm robot more resembles a human operator, and the human experience and knowledge can be transferred and mimicked more easily [14,82,192]. In order the dual-arm robots to implement assembly tasks, some involved issues have been addressed, such as the scheduling and workspace layout design, environment state perception, optimization for motion planning, and operational space path planning etc. [193–196]. However, currently there are still plenty of challenges to explore, e.g., the control model of extensively increased DOFs (maybe exceeds 50-DOF for an upper-body robot), path planning and collision avoidance, dynamic adjustment of objects, and usage of various assembly tools. Furthermore, current exploration manners, perception mechanism and manipulation processes for single-arm robots will become infeasible in dual-arm case, and dynamic flexible collaborations of dual arms and hands should be taken into full consideration in constructing the perception and control models.

Authors' contributions

Longhui Qin confirms sole responsibility for the following: study conception and design, data collection, analysis and interpretation of results, and manuscript preparation. All authors read and approved the final manuscript.

Funding

Supported by National Natural Science Foundation of China (Grant No. 52205009), Open Foundation of State Key Laboratory of Mechanical Transmission for Advanced Equipment of Chongqing University (Grant No. SKLMT-MSKFKT-202307), and Taihu Lake Innovation Fund for the School of Future Technology of Southeast University.

Declaration of Competing Interest

No competing interests should be disclosed.

Acknowledgements

Not applicable.

References

[1] Andrzejewski K, Cooper M, Griffiths C, et al. Optimisation process for robotic assembly of electronic components. Int J Adv Manuf Technol 2018;99:2523–35.
[2] Kunic A, Naboni R, Kramberger A, et al. Design and assembly automation of the robotic reversible timber beam. Autom Constr 2021;123:103531.
[3] An Y, Yang J, He B, et al. A microscopic vision-based robotic system for floating electrode assembly. IEEE/ASME Trans Mechatron 2024;29(5):3810–20.
[4] Fernandez SR, Olabi A, Gibaru O. Multi-surface admittance control approach applied on robotic assembly of large-scale parts in aerospace manufacturing. 19th International Conference on Advanced Robotics (ICAR) IEEE; 2019. p. 688–94.
[5] Li R, Qiao H. A survey of methods and strategies for High-Precision robotic grasping and assembly tasks — some new trends. IEEE/ASME Trans Mechatron 2019;24(6):2718–32.
[6] Jiang J, Yao L, Huang Z, et al. The state of the art of search strategies in robotic assembly. J Ind Inf Integr 2022;26:100259.
[7] Jiang Y, Huang Z, Yang B, et al. A review of robotic assembly strategies for the full operation procedure: planning, execution and evaluation. Rob Comput Integr Manuf 2022;78:102366.
[8] Xu J, Zhang C, Liu Z, et al. A review on significant technologies related to the robot-guided intelligent bolt assembly under complex or uncertain working conditions. IEEE Access 2019;7:136752–76.
[9] Navas-Reascos GE, Romero D, Stahre J, et al. Wire harness assembly process supported by collaborative robots: literature review and call for R&D. Robotics 2022;11(3):65.
[10] Kiyokawa T, Shirakura N, Wang Z, et al. Difficulty and complexity definitions for assembly task allocation and assignment in human–robot collaborations: a review. Rob Comput Integr Manuf 2023;84:102598.
[11] Monguzzi A, Pelosi M, Zanchettin AM, et al. Tactile based robotic skills for cable routing operations. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2023. p. 3793–9.
[12] Royo-Miquel J, Hamaya M, Beltran-Hernandez CC, et al. Learning robotic assembly by leveraging physical softness and tactile sensing. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2023. p. 6469–76.
[13] Schmitt R, Cai Y. Single camera-based synchronisation within a concept of robotic assembly in motion. Assem Autom 2014;34(2):160–8.
[14] Huang Y, Zhang X, Chen X, et al. Vision-guided peg-in-hole assembly by baxter robot. Adv Mech Eng 2017;9(12):1687814017748078.
[15] Lin HI. Development of an intelligent transformer insertion system using a robot arm. Rob Comput Integr Manuf 2018;51:209–21.
[16] Xu R, Zhao X, Liu F, et al. High-precision monocular vision guided robotic assembly based on local pose invariance. IEEE Trans Instrum Meas 2023;72:1–12.
[17] Chen J, Liu Z, Xu J, et al. A novel disassembly strategy of hexagonal screws based on robot vision and robot-tool cooperated motion. Appl Sci 2022;13(1):251.
[18] Chang WC. Robotic assembly of smartphone back shells with eye-in-hand visual servoing. Rob Comput Integr Manuf 2018;50:102–13.
[19] Li D, Mu Q, Yuan Y, et al. 6D pose estimation based on 3D edge binocular reprojection optimization for robotic assembly. IEEE Robot Autom Lett 2023;8(12):8319–26.
[20] Bartyzel G, Półchłopek W, Rzepka D. Reinforcement learning with stereo-view observation for robust electronic component robotic insertion. J Intell Robot Syst 2023;109(3):57.
[21] Choi C, Rus D. Probabilistic visual verification for robotic assembly manipulation. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2016. p. 5656–63.

[22] Litvak Y, Biess A, Bar-Hillel A. Learning pose estimation for high-precision robotic assembly using simulated depth images. International Conference on Robotics and Automation (ICRA 2019. IEEE; 2019. p. 3521–7.
[23] Waltersson GA, Laezza R, Karayiannidis Y. Planning and control for cable-routing with dual-arm robot. International Conference on Robotics and Automation (ICRA 2022. IEEE; 2022. p. 1046–52.
[24] Song R, Li F, Fu T, et al. A robotic automatic assembly system based on vision. Appl Sci 2020;10(3):1157.
[25] Jiang T, Cui H, Cheng X, et al. A measurement method for robot peg-in-hole prealignment based on combined two-level visual sensors. IEEE Trans Instrum Meas 2020;70:1–12.
[26] Xu J, Liu K, Pei Y, et al. A noncontact control strategy for circular peg-in-hole assembly guided by the 6-dof robot based on hybrid vision. IEEE Trans Instrum Meas 2022;71:1–15.
[27] Dharmara K, Monfared RP, Ogun PS, et al. Robotic assembly of threaded fasteners in a non-structured environment. Int J Adv Manuf Technol 2018;98:2093–107.
[28] Kang H, Zang Y, Wang X, et al. Uncertainty-driven spiral trajectory for robotic peg-in-hole assembly. IEEE Robot Autom Lett 2022;7(3):6661–8.
[29] Huang H, Cheng H, Wang T, et al. Hole-finding learning strategy for a robot assembly with keyed circular peg. IEEE Access 2023;11:21488–97.
[30] Zhang X, Eltouny K, Liang X, et al. Automatic screw detection and tool recommendation system for robotic disassembly. J Manuf Sci Eng 2023;145(3):031008.
[31] Fu B, Leong SK, Lian X, et al. 6D robotic assembly based on RGB-only object pose estimation. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2022. p. 4736–42.
[32] Mou F, Ren H, Wang B, et al. Pose estimation and robotic insertion tasks based on YOLO and layout features. Eng Appl Artif Intell 2022;114:105164.
[33] Liu Y, Zhou J, Li Y, et al. A high-accuracy pose measurement system for robotic automated assembly in large-scale space. Measurement 2022;188:110426.
[34] von Drigalski F, Hayashi K, Huang Y, et al. Precise multi-modal in-hand pose estimation using low-precision sensors for robotic assembly. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2021. p. 968–74.
[35] Jiang Y, Liu G, Huang Z, et al. Geometry perception and motion planning in robotic assembly based on semantic segmentation and point clouds reconstruction. Eng Appl Artif Intell 2024;130:107678.
[36] André AN, Lehmann O, Govilas J, et al. Automating robotic micro-assembly of fluidic chips and single fiber compression tests based-on Θ visual measurement with high-precision fiducial markers. IEEE Trans Autom Sci Eng 2022;21(1):353–66.
[37] Mei B, Liang Z, Xie Y, et al. Positioning accuracy enhancement of a robotic assembly system for thin-walled aerostructure assembly. J Ind Inf Integr 2023;35:100518.
[38] Hao T, Xu D. Robotic grasping and assembly of screws based on visual servoing using point features. Int J Adv Manuf Technol 2023;129(9):3979–91.
[39] Stolt A, Linderoth M, Robertsson A, et al. Detection of contact force transients in robotic assembly. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2015. p. 962–8.
[40] Roberge E, Duchaine V. Detecting insertion tasks using convolutional neural networks during robot teaching-by-demonstration. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2017. p. 3210–6.
[41] Salem A, Karayiannidis Y. Robotic assembly of rounded parts with and without threads. IEEE Robot Autom Lett 2020;5(2):2467–74.
[42] Gai Y, Wang B, Zhang J, et al. Piecewise strategy and decoupling control method for high pose precision robotic peg-in-hole assembly. Rob Comput Integr Manuf 2023;79:102451.
[43] Choi S, Kim D, Kim Y, et al. A novel compliance compensator capable of measuring six-axis force/torque and displacement for a robotic assembly. IEEE/ASME Trans Mechatron 2023;29(1):29–40.
[44] Zhang S, Jiang Q, Li Y, et al. Contact state classification in industrial robotic assembly tasks based on extreme learning machine. IEEE 8th Annual International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER) IEEE; 2018. p. 617–22.
[45] Jasim IF, Plapper PW. Contact-state monitoring of force-guided robotic assembly tasks using expectation maximization-based Gaussian mixtures models. Int J Adv Manuf Technol 2014;73:623–33.
[46] Jasim IF, Plapper PW, Voos H. Contact-state modelling in force-controlled robotic peg-in-hole assembly processes of flexible objects using optimised Gaussian mixtures. Proc Inst Mech Eng Part B J Eng Manuf 2017;231(8):1448–63.
[47] Romeres D, Jha DK, Yerazunis W, et al. Anomaly detection for insertion tasks in robotic assembly using Gaussian process models. 18th European Control Conference (ECC) IEEE; 2019. p. 1017–22.
[48] Halt L, Pan F, Tenbrock P, et al. A transferable force controller based on prescribed performance for contact establishment in robotic assembly tasks. IEEE 15th International Conference on Automation Science and Engineering (CASE) IEEE; 2019. p. 830–5.
[49] Almeida D, Karayiannidis Y. Folding assembly by means of dual-arm robotic manipulation. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2016. p. 3987–93.
[50] Li D, Zhao H, Ge D, et al. A novel robotic multiple peg-in-hole assembly pipeline: modeling, strategy, and control. IEEE/ASME Trans Mechatron 2023.
[51] Polverini MP, Zanchettin AM, Castello S, et al. Sensorless and constraint based peg-in-hole task execution with a dual-arm robot. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2016. p. 415–20.
[52] Park H, Park J, Lee DH, et al. Compliance-based robotic peg-in-hole assembly strategy without force feedback. IEEE Trans Ind Electron 2017;64(8):6299–309.
[53] Zhang T, Liang X, Zou Y. Robot peg-in-hole assembly based on contact force estimation compensated by convolutional neural network. Control Eng Pr 2022;120:105012.
[54] Wang C, Luo H, Zhang K, et al. POMDP-Guided active Force-Based search for robotic insertion. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2023. p. 10668–75.
[55] Grella F, Baldini G, Canale R, et al. A tactile sensor-based architecture for collaborative assembly tasks with heavy-duty robots. 20th International Conference on Advanced Robotics (ICAR) IEEE; 2021. p. 1030–5.
[56] Ishige M, Umedachi T, Ijiri Y, et al. Blind bin picking of small screws through in-finger manipulation with compliant robotic fingers. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2020. p. 9337–44.
[57] Caccavale R, Finzi A, Laudante G, et al. Manipulation of boltlike fasteners through fingertip tactile perception in robotic assembly. IEEE/ASME Trans Mechatron 2023;29(2):820–31.
[58] Nozu K, Shimonomura K. Robotic bolt insertion and tightening based on in-hand object localization and force sensing. IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM) IEEE; 2018. p. 310–5.
[59] Wilson A, Jiang H, Lian W, et al. Cable routing and assembly using tactile-driven motion primitives. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2023. p. 10408–14.
[60] Hu Z, Lin L, Lin W, et al. Machine learning for tactile perception: advancements, challenges, and opportunities. Adv Intell Syst 2023;5(7):2200371.
[61] Li T, Yan Y, Yu C, et al. A comprehensive review of robot intelligent grasping based on tactile perception. Rob Comput Integr Manuf 2024;90:102792.
[62] Park H, Gbadam GS, Niu S, et al. Manufacturing strategies for highly sensitive and self-powered piezoelectric and triboelectric tactile sensors. Int J Extrem Manuf 2024;7(1):012006.
[63] Kong H, Li W, Song Z, et al. Recent advances in multimodal sensing integration and decoupling strategies for tactile perception. Mater Futures 2024;3(2):022501.
[64] Sun Y, Cheng N, Zhang S, et al. Tactile data generation and applications based on visuo-tactile sensors: a review. Inf Fusion 2025:103162.
[65] Von Drigalski F, Nakashima C, Shibata Y, et al. Team O2AS at the world robot summit 2018: an approach to robotic kitting and assembly tasks using general purpose grippers and tools. Adv Rob 2020;34(7-8):514–30.
[66] Ma Y, Du K, Zhou D, et al. Automatic precision robot assembly system with microscopic vision and force sensor. Int J Adv Rob Syst 2019;16(3):1729881419851619.
[67] Zhang Z, Wang Y, Zhang Z, et al. A residual reinforcement learning method for robotic assembly using visual and force information. J Manuf Syst 2024;72:245–62.
[68] De Gregorio D, Zanella R, Palli G, et al. Integration of robotic vision and tactile sensing for wire-terminal insertion tasks. IEEE Trans Autom Sci Eng 2018;16(2):585–98.
[69] Li S, Gu H. Acoustic contacting detection in robotic accurate assembly. IEEE international conference on robotics and biomimetics (ROBIO) IEEE; 2017. p. 1829–32.
[70] Wojciechowski J, Suszynski M. Optical scanner assisted robotic assembly. Assem Autom 2017;37(4):434–41.
[71] Liu Z, Xie Y, Xu J, et al. Laser tracker based robotic assembly system for large scale peg-hole parts. The 4th Annual IEEE International Conference on Cyber Technology in Automation, Control and Intelligent. IEEE; 2014. p. 574–8.
[72] Qin Z, Wang P, Sun J, et al. Precise robotic assembly for large-scale objects based on automatic guidance and alignment. IEEE Trans Instrum Meas 2016;65(6):1398–411.
[73] Harada K, Nagata K, Rojas J, et al. Proposal of a shape adaptive gripper for robotic assembly tasks. Adv Rob 2016;30(17-18):1186–98.
[74] Van Wyk K, Culleton M, Falco J, et al. Comparative peg-in-hole testing of a force-based manipulation controlled robotic hand. IEEE Trans Robot 2018;34(2):542–9.
[75] Tao R, Fan J, Jing F, et al. A robotic end-effector for screwing and unscrewing bolts from the side. IEEE Robot Autom Lett 2022;7(4):9786–93.
[76] Hammond M, Dempsey A, Ward W, et al. A hybrid soft material robotic end-effector for reversible in-space assembly of strut components. Front Robot AI 2023;10:1099297.
[77] Polishchuk M, Telenyk S, Tkach M. Adaptive assembly module for industrial robot: design and simulation. FME Trans 2022;50(1).
[78] Hartisch RM, Haninger K. Compliant finray-effect gripper for high-speed robotic assembly of electrical components. IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM) IEEE; 2023. p. 375–80.
[79] Shibata K, Dobashi H. Development of a versatile robotic hand toward jig-less assembly of a shaft-shaped part. IEEE Robot Autom Lett 2023;9(2):1222–9.
[80] Hashimoto Y, Yatou D, Yamada T, et al. Fundamental study on robotic assembly of modular fixture parts by a low cost 4-finger 12 dof hand. IEEE/SICE International Symposium on System Integration (SII) IEEE; 2015. p. 930–5.
[81] Shimada F, Murakami K, Senoo T, et al. Bolt loosening detection using multi-purpose robot hand. IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM) IEEE; 2020. p. 1860–6.
[82] Lee DH, Choi MS, Park H, et al. Peg-in-hole assembly with dual-arm robot and dexterous robot hands. IEEE Robot Autom Lett 2022;7(4):8566–73.
[83] Nie K, Wan W, Harada K. An adaptive robotic gripper with L-shape fingers for peg-in-hole tasks. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2018. p. 4022–8.
[84] Su S, Pham DT, Ji C, et al. Design of a compliant device for peg-hole separation in robotic disassembly. Int J Adv Manuf Technol 2023;124(9):3011–9.
[85] Gilday K, Hughes J, Iida F. Gel-based soft interfaces for fastener manipulation in robotic agile assembly. 2019 2nd IEEE International Conference on Soft Robotics (RoboSoft). IEEE; 2019. p. 155–60.

[86] Sun X, Wang C, Chen W, et al. A single-actuator four-finger adaptive gripper for robotic assembly. IEEE Trans Ind Electron 2023;70(12):12555–65.
[87] Zhao J, Wang X, Wang S, et al. Assembly of randomly placed parts realized by using only one robot arm with a general parallel-jaw gripper. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2020. p. 5024–30.
[88] Peng YC, Jivani D, Radke RJ, et al. Comparing position-and image-based visual servoing for robotic assembly of large structures. IEEE 16th International Conference on Automation Science and Engineering (CASE) IEEE; 2020. p. 1608–13.
[89] Ma Y, Liu X, Zhang J, et al. Robotic grasping and alignment for small size components assembly based on visual servoing. Int J Adv Manuf Technol 2020;106:4827–43.
[90] Yan S, Tao X, Xu D. High-precision robotic assembly system using three-dimensional vision. Int J Adv Rob Syst 2021;18(3):17298814211027029.
[91] Liu Z, Wang K, Liu D, et al. A motion planning method for visual servoing using deep reinforcement learning in autonomous robotic assembly. IEEE/ASME Trans Mechatron 2023;28(6):3513–24.
[92] Jha DK, Romeres D, Jain S, et al. Design of adaptive compliance controllers for safe robotic assembly. European Control Conference (ECC) IEEE; 2023. p. 1–8.
[93] Sheng J, Tang Y, Xu S, et al. A stable learning-based method for robotic assembly with motion and force measurements. IEEE Trans Ind Electron 2023;71(9):11093–103.
[94] Luo J, Solowjow E, Wen C, et al. Deep reinforcement learning for robotic assembly of mixed deformable and rigid objects. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2018. p. 2062–9.
[95] Jiao C, Zhang L, Su X, et al. Task-based compliance control for bottle screw manipulation with a dual-arm robot. IEEE Trans Ind Electron 2023;71(2):1823–31.
[96] Li B, Chen H, Jin T. Support vector regression for optimal robotic force control assembly. J Manuf Sci Eng 2020;142(1):011007.
[97] Chen Z, Xie S, Zhang X. Position/force visual-sensing-based robotic sheet-like peg-in-hole assembly. IEEE Trans Instrum Meas 2021;71:1–11.
[98] Peng YC, Chen S, Jivani D, et al. Sensor-guided assembly of segmented structures with industrial robots. Appl Sci 2021;11(6):2669.
[99] Qiao H, Wang M, Su J, et al. The concept of "attractive region in environment" and its application in high-precision tasks with low-precision systems. IEEE/ASME Trans Mechatron 2014;20(5):2311–27.
[100] Li X, Li R, Qiao H, et al. Human-inspired compliant strategy for peg-in-hole assembly using environmental constraint and coarse force information. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2017. p. 4743–8.
[101] Manyar OM, Narayan SV, Lengade R, et al. Physics-informed learning to enable robotic screw-driving under hole pose uncertainties. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2023. p. 2993–3000.
[102] Zhang Y, Koga Y, Balkcom D. Interlocking block assembly with robots. IEEE Trans Autom Sci Eng 2021;18(3):902–16.
[103] Kim CH, Seo J. Shallow-depth insertion: peg in shallow hole through robotic in-hand manipulation. IEEE Robot Autom Lett 2019;4(2):383–90.
[104] Mobedi E, Villa N, Kim W, et al. An adaptive control approach to robotic assembly with uncertainties in vision and dynamics. 29th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN) IEEE; 2020. p. 144–50.
[105] Su J, Liu C, Li R. Robot precision assembly combining with passive and active compliant motions. IEEE Trans Ind Electron 2021;69(8):8157–67.
[106] Cheng H, Liu T, Zhang W, et al. Robotic assembly for tight tolerant cascaded peg-in-hole process with incomplete constraints. Assem Autom 2020;40(5):769–78.
[107] Gorjup G, Gao G, Dwivedi A, et al. A flexible robotic assembly system combining cad based localization, compliance control, and a multi-modal gripper. IEEE Robot Autom Lett 2021;6(4):8639–46.
[108] Friedrich C, Lechler A, Verl A. The control architecture KaReSA: a learning disassembly system. IEEE International Conference on Industrial Technology (ICIT) IEEE; 2016. p. 113–8.
[109] Pane Y, Arbo MH, Aertbeliën E, et al. A system architecture for cad-based robotic assembly with sensor-based skills. IEEE Trans Autom Sci Eng 2020;17(3):1237–49.
[110] Koga Y, Kerrick H, Chitta S. On cad informed adaptive robotic assembly. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2022. p. 10207–14.
[111] Chervinskii F, Zobov S, Rybnikov A, et al. Auto-Assembly: a framework for automated robotic assembly directly from CAD. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2023. p. 11294–300.
[112] Navarro-Gonzalez J, Lopez-Juarez I, Rios-Cabrera R, et al. On-line knowledge acquisition and enhancement in robotic assembly tasks. Rob Comput Integr Manuf 2015;33:78–89.
[113] Nottensteiner K, Sagardia M, Stemmer A, et al. Narrow passage sampling in the observation of robotic assembly tasks. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2016. p. 130–7.
[114] Nägele F, Halt L, Tenbrock P, et al. A prototype-based skill model for specifying robotic assembly tasks. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2018. p. 558–65.
[115] Nottensteiner K, Stulp F, Albu-Schäffer A. Robust, locally guided peg-in-hole using impedance-controlled robots. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2020. p. 5771–7.
[116] Liu Q, Ji Z, Xu W, et al. Knowledge-guided robot learning on compliance control for robotic assembly task with predictive model. Expert Syst Appl 2023;234:121037.
[117] Man Z, Fengming L, Wei Q, et al. Robot bolt skill learning based on GMM-GMR. International Conference on Intelligent Robotics and Applications. Springer; 2021. p. 235–45.
[118] Cao Z, Hu H, Yang X, et al. A robot 3C assembly skill learning method by intuitive human assembly demonstration. WRC Symposium on Advanced Robotics and Automation (WRC SARA) IEEE; 2019. p. 13–8.
[119] Hamaya M, von, Drigalski F, Matsubara T, et al. Learning soft robotic assembly strategies from successful and failed demonstrations. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2020. p. 8309–15.
[120] Kyrarini M, Haseeb MA, Ristić-Durrant D, et al. Robot learning of industrial assembly task via human demonstrations. Auton Robots 2019;43:239–57.
[121] Lin H-I. Design of an intelligent robotic precise assembly system for rapid teaching and admittance control. Rob Comput Integr Manuf 2020;64:101946.
[122] Song J, Chen Q, Li Z. A peg-in-hole robot assembly system based on gauss mixture model. Rob Comput Integr Manuf 2021;67:101996.
[123] Qin F, Xu D, Zhang D, et al. Robotic skill learning for precision assembly with microscopic vision and force feedback. IEEE/ASME Trans Mechatron 2019;24(3):1117–28.
[124] Zhang S, Huang H, Huang D, et al. Subtask-learning based for robot self-assembly in flexible collaborative assembly in manufacturing. Int J Adv Manuf Technol 2022;120(9):6807–19.
[125] Tang T, Lin HC, Zhao Y, et al. Teach industrial robots peg-hole-insertion by human demonstration. IEEE International Conference on Advanced Intelligent Mechatronics (AIM) IEEE; 2016. p. 488–94.
[126] Hu H, Yang X, Lou Y. A robot learning from demonstration framework for skillful small parts assembly. Int J Adv Manuf Technol 2022;119(9):6775–87.
[127] Mendes N, Neto P, Safeea M, et al. Online robot teleoperation using human hand gestures: a case study for assembly operation. Robot 2015: Second Iberian Robotics Conference: Advances in Robotics 2. Springer; 2016. p. 93–104.
[128] Liu Y, Er MJ, Guo C. Online time-optimal path and trajectory planning for robotic multipoint assembly. Assem Autom 2021;41(5):601–11.
[129] Li S, Yuan X, Niu J. Robotic peg-in-hole assembly strategy research based on reinforcement learning algorithm. Appl Sci 2022;12(21):11149.
[130] Hou Z, Fei J, Deng Y, et al. Data-efficient hierarchical reinforcement learning for robotic assembly control applications. IEEE Trans Ind Electron 2020;68(11):11565–75.
[131] Eteke C, Kebüde D, Akgün B. Reward learning from very few demonstrations. IEEE Trans Robot 2020;37(3):893–904.
[132] Roveda L, Magni M, Cantoni M, et al. Assembly task learning and optimization through Human's demonstration and machine learning. IEEE International Conference on Systems, Man, and Cybernetics (SMC) IEEE; 2020. p. 1852–9.
[133] Güleçyüz B, von Büren V, Xu X, et al. Netlfd: network-aware learning from demonstration for in-contact skills via teleoperation. IEEE Robot Autom Lett 2023;8(11):6995–7002.
[134] Huang I, Bajcsy R. Robot learning from demonstration with tactile signals for geometry-dependent tasks. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2020. p. 8323–8.
[135] Saveriano M, Abu-Dakka FJ, Kramberger A, et al. Dynamic movement primitives in robotics: a tutorial survey. Int J Robot Res 2023;42(13):1133–84.
[136] Savarimuthu TR, Buch AG, Schlette C, et al. Teaching a robot the semantics of assembly tasks. IEEE Trans Syst Man Cyber Syst 2017;48(5):670–92.
[137] Tunstel E, Dani A, Martinez C, et al. Robotic wire pinning for wire harness assembly automation. IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM) IEEE; 2020. p. 1208–15.
[138] Ding G, Liu Y, Zang X, et al. A task-learning strategy for robotic assembly tasks from human demonstrations. Sensors 2020;20(19):5505.
[139] Zhao H, Chen Y, Li X, et al. Robotic peg-in-hole assembly based on reversible dynamic movement primitives and trajectory optimization. Mechatronics 2023;95:103054.
[140] Iturrate I, Roberge E, Østergaard EH, et al. Improving the generalizability of robot assembly tasks learned from demonstration via CNN-based segmentation. IEEE 15th International Conference on Automation Science and Engineering (CASE) IEEE; 2019. p. 553–60.
[141] Scherzinger S, Roennau A, Dillmann R. Contact skill imitation learning for robot-independent assembly programming. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2019. p. 4309–16.
[142] Ji S, Lee S, Yoo S, et al. Learning-based automation of robotic assembly for smart manufacturing. Proc IEEE 2021;109(4):423–40.
[143] Wang K, Zhao Y, Sakuma I. Learning robotic insertion tasks from human demonstration. IEEE Robot Autom Lett 2023;8(9):5815–22.
[144] Raessa M, Chen JCY, Wan W, et al. Human-in-the-loop robotic manipulation planning for collaborative assembly. IEEE Trans Autom Sci Eng 2020;17(4):1800–13.
[145] Elguea-Aguinaco Í, Serrano-Muñoz A, Chrysostomou D, et al. A review on reinforcement learning for contact-rich robotic manipulation tasks. Rob Comput Integr Manuf 2023;81:102517.
[146] Levine S, Abbeel P. Learning neural network policies with guided policy search under unknown dynamics. NIPS'14: Proceedings of the 28th International Conference on Neural Information Processing Systems. Cambridge, USA: MIT Press; 2014. p. 1071–9.
[147] Thomas G, Chien M, Tamar A, et al. Learning robotic assembly from cad. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2018. p. 3524–31.
[148] Luo J, Solowjow E, Wen C, et al. Reinforcement learning on variable impedance controller for high-precision robotic assembly. International Conference on Robotics and Automation (ICRA) IEEE; 2019. p. 3080–7.
[149] Liu D, Zhang X, Du Y, et al. Industrial insert robotic assembly based on model-based meta-reinforcement learning. IEEE International Conference on Robotics and Biomimetics (ROBIO) IEEE; 2021. p. 1508–12.

[150] Li F, Jiang Q, Zhang S, et al. Robot skill acquisition in assembly process using deep reinforcement learning. Neurocomputing 2019;345:92–102.
[151] Song R, Li F, Quan W, et al. Skill learning for robotic assembly based on visual perspectives and force sensing. Rob Auton Syst 2021;135:103651.
[152] Oikawa M, Kusakabe T, Kutsuzawa K, et al. Reinforcement learning for robotic assembly using non-diagonal stiffness matrix. IEEE Robot Autom Lett 2021;6(2):2737–44.
[153] Chen C, Zhang C, Pan Y. Active compliance control of robot peg-in-hole assembly based on combined reinforcement learning. Appl Intell 2023;53(24):30677–90.
[154] Johnson D, Chen G, Lu Y. Multi-agent reinforcement learning for real-time dynamic production scheduling in a robot assembly cell. IEEE Robot Autom Lett 2022;7(3):7684–91.
[155] Zhang X, Jin S, Wang C, et al. Learning insertion primitives with discrete-continuous hybrid action space for robotic assembly tasks. International conference on robotics and automation (ICRA) IEEE; 2022. p. 9881–7.
[156] Lin N, Li Y, Tang K, et al. Manipulation planning from demonstration via goal-conditioned prior action primitive decomposition and alignment. IEEE Robot Autom Lett 2022;7(2):1387–94.
[157] Khader SA, Yin H, Falco P, et al. Stability-guaranteed reinforcement learning for contact-rich manipulation. IEEE Robot Autom Lett 2020;6(1):1–8.
[158] Langaa J, Sloth C. Expert initialized reinforcement learning with application to robotic assembly. IEEE 18th International Conference on Automation Science and Engineering (CASE) IEEE; 2022. p. 1405–10.
[159] Beltran-Hernandez CC, Petit D, Ramirez-Alpizar IG, et al. Variable compliance control for robotic peg-in-hole assembly: a deep-reinforcement-learning approach. Appl Sci 2020;10(19):6923.
[160] Ahn KH, Na M, Song JB. Robotic assembly strategy via reinforcement learning based on force and visual information. Rob Auton Syst 2023;164:104399.
[161] Wang Y, Beltran-Hernandez CC, Wan W, et al. Robotic imitation of human assembly skills using hybrid trajectory and force learning. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2021. p. 11278–84.
[162] Deng Y., Hou Z., Yang W., et al. Sample-efficiency, stability and generalization analysis for deep reinforcement learning on robotic peg-in-hole assembly. Intelligent Robotics and Applications: 14th International Conference, ICIRA 2021, Yantai, China, October 22–25, Proceedings, Part II 14, Springer; 2021, p. 393–403.
[163] Wang X, Wang S, Liang X, et al. Deep reinforcement learning: a survey. IEEE Trans Neural Netw Learn Syst 2022;354(4):5064–78.
[164] Li F, Jiang Q, Quan W, et al. Manipulation skill acquisition for robotic assembly using deep reinforcement learning. IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM) IEEE; 2019. p. 13–8.
[165] Apolinarska AA, Pacher M, Li H, et al. Robotic assembly of timber joints using reinforcement learning. Autom Constr 2021;125:103569.
[166] Luo J, Li H. A learning approach to robot-agnostic force-guided high precision assembly. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2021. p. 2151–7.
[167] Xu J, Hou Z, Wang W, et al. Feedback deep deterministic policy gradient with fuzzy reward for robotic multiple peg-in-hole assembly tasks. IEEE Trans Ind Inf 2018;15(3):1658–67.
[168] Hou Z, Li Z, Hsu C, et al. Fuzzy logic-driven variable time-scale prediction-based reinforcement learning for robotic multiple peg-in-hole assembly. IEEE Trans Autom Sci Eng 2020;19(1):218–29.
[169] Leyendecker L, Schmitz M, Zhou HA, et al. Deep reinforcement learning for robotic control in high-dexterity assembly tasks – reward curriculum approach. Int J Semant Comput 2022;16(3):381–402.
[170] Chen C, Zhang H, Pan Y, et al. Robot autonomous grasping and assembly skill learning based on deep reinforcement learning. Int J Adv Manuf Technol 2024;130(11):5233–49.
[171] Xiao R, Yang C, Jiang Y, et al. One-shot sim-to-real transfer policy for robotic assembly via reinforcement learning with visual demonstration. Robotica 2024;42(4):1074–93.
[172] Chen W, Zeng C, Liang H, et al. Multimodality driven impedance-based sim2real transfer learning for robotic multiple peg-in-hole assembly. IEEE Trans Cyber 2023;54(5):2784–97.
[173] Vuong N, Pham QC. Contact reduction with bounded stiffness for robust Sim-to-Real transfer of robot assembly. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) IEEE; 2023. p. 361–7.
[174] Liu N, Cai Y, Lu T, et al. Real–sim–real transfer for real-world robot control policy learning with deep reinforcement learning. Appl Sci 2020;10(5):1555.
[175] Chen Y, Li X, Guo S, et al. Real2sim or sim2real: robotics visual insertion using deep reinforcement learning and real2sim policy adaptation. International Conference on Intelligent Autonomous Systems. Springer; 2022. p. 617–29.
[176] Qian K, Duan Y, Luo C, et al. Pixel-level domain adaptation for real-to-sim object pose estimation. IEEE Trans Cogn Dev Syst 2023;15(3):1618–27.
[177] Wang L, Guo R, Vuong Q, et al. A real2sim2real method for robust object grasping with neural surface reconstruction. IEEE 19th International Conference on Automation Science and Engineering (CASE) IEEE; 2023. p. 1–8.
[178] Tanaka K, Yonetani R, Hamaya M, et al. Trans-am: transfer learning by aggregating dynamics models for soft robotic assembly. IEEE International Conference on Robotics and Automation (ICRA) IEEE; 2021. p. 4627–33.
[179] Men Y, Jin L, Cui T, et al. Policy fusion transfer: the knowledge transfer for different robot peg-in-hole insertion assemblies. IEEE Trans Instrum Meas 2023;72:1–10.
[180] Jin L, Men Y, Song R, et al. Robot skill generalization: feature-selected adaptation transfer for peg-in-hole assembly. IEEE Trans Ind Electron 2023;71(3):2748–57.
[181] Kluz R, Trzepieciński T. Analysis of the optimal orientation of robot gripper for an improved capability assembly process. Rob Auton Syst 2015;74:253–66.
[182] Shen L, Su J, Zhang X. Review on peg-in-hole insertion technology based on reinforcement learning. China Automation Congress (CAC) IEEE; 2023. p. 6688–95.
[183] Jin S, Lian W, Wang C, et al. Robotic cable routing with spatial representation. IEEE Robot Autom Lett 2022;7(2):5687–94.
[184] Shi X, Wang Y, Qin L. Surface recognition with a bioinspired tactile fingertip. IEEE Sens J 2023;23(16):18842–55.
[185] Lepora NF. The future lies in a pair of tactile hands. Sci Robot 2024;9(91):eadq1501.
[186] Qin L, Shi X, Yang W, et al. Surface recognition with a tactile finger based on automatic features transferred from deep learning. IEEE Trans Instrum Meas 2024;73:1–10.
[187] Kristiansen E, Nielsen EK, Hansen L, et al. A novel strategy for automatic error classification and error recovery for robotic assembly in flexible production. J Intell Robot Syst 2020;100(3):863–77.
[188] Chen F, Cannella F, Huang J, et al. A study on error recovery search strategies of electronic connector mating for robotic fault-tolerant assembly. J Intell Robot Syst 2016;81:257–71.
[189] Laursen JS, Ellekilde LP, Schultz UP. Modelling reversible execution of robotic assembly. Robotica 2018;36(5):625–54.
[190] Li R, Pham DT, Huang J, et al. Unfastening of hexagonal headed screws by a collaborative robot. IEEE Trans Autom Sci Eng 2020;17(3):1455–68.
[191] Dogar M, Spielberg A, Baker S, et al. Multi-robot grasp planning for sequential assembly operations. Auton Robots 2019;43:649–64.
[192] Fang S, Huang X, Chen H, et al. Dual-arm robot assembly system for 3C product based on vision guidance. IEEE International Conference on Robotics and Biomimetics (ROBIO) IEEE; 2016. p. 807–12.
[193] Wessén J, Carlsson M, Schulte C, et al. A constraint programming model for the scheduling and workspace layout design of a dual-arm multi-tool assembly robot. Constraints 2023;28(2):71–104.
[194] Wang Z, Gan Y, Dai X. An environment state perception method based on knowledge representation in dual-arm robot assembly tasks. Int J Intell Robot Appl 2020;4(2):177–90.
[195] Ying K-C, Pourhejazy P, Cheng CY, et al. Deep learning-based optimization for motion planning of dual-arm assembly robots. Comput Ind Eng 2021;160:107603.
[196] Lim S-J, Han C-S. Operational space path planning of the dual-arm robot for the assembly task. Int J Precis Eng Manuf 2014;15(10):2071–6.

Longhui Qin received his bachelor's degree in mechanical engineering and automation from *Xi'an Jiaotong University, China,* in 2011. Then he received the Ph.D. degree in mechatronic engineering from *Zhejiang University, China,* in 2016. From 2017–2020, he worked as a research fellow and postdoc scholar at *Nanyang Technological University, Singapore,* and *California University Los Angeles, USA,* respectively. Currently, he works as an associate professor at *Southeast University, China*. His research fields mainly include tactile sensors, computer vision, dexterous manipulation, mechanics, and human-machine interaction, etc.